picBunch v0.7.0: a free-space FFT convolution for the initial field of a uniformly moving particle bunch, and its measured cost and accuracy

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Release v0.7.0 (commit a152c518, tag 25d6ca8b), run pr24-20261102

1 Introduction

When a particle-in-cell (PIC) simulation starts, the bunch of particles must carry an initial electromagnetic field, or the first push already acts on the wrong force. The field of a moving bunch in free space is known in closed form for each point charge, but a simulation reads that field on a staggered mesh with thousands of observation points, so it is computed by a fast sum over the source charges rather than by evaluating the closed form everywhere. We compute it as a free-space FFT convolution of a TSC deposit with the sampled TSC-convolved point-charge kernel. The field is the present-time field of uniformly moving charges, with no radiation term and no history. This paper gives the exact discrete recipe a re-implementer needs, and it measures the cost and the accuracy of picBunch v0.7.0 (commit a152c518, tag 25d6ca8b) on a V100, an H100 and a gfx1100, naming the reference for every error statement. The release is driven by a single TOML configuration (input file, destination, config; Section 3.1), and it carries the capabilities the measurements below exercise: an opt-in near shell with a fast closed-form evaluator, MPI component-parallel mesh and tile-parallel near shell with bitwise equality across rank counts, a tolerance-derived far-tail margin for zero fill, and a float32 checkpoint that restarts under a stock PIConGPU.

The established workaround avoids computing the initial field altogether: one starts from a field-free state and marches the bunch forward, letting the fields build up in the first few steps. The present-time field of a uniformly moving bunch has no transient, so this history has no physical basis. We measure how well the discrete method reproduces that field and how much it costs, and we state the conditions under which the measured error is largest (Section 5).

2 Field definition

The computed quantity is a single linear integral. For a bunch of source charges at positions $\mathbf{x}_p$ with signed charge q_p, and a shared dimensionless velocity $\mathbf{u} = \mathbf{p}/(mc)$, the field at a staggered observation point $\mathbf{x}_{\mathrm{obs}}$ is the TSC-convolved uniformly-moving point-charge field

$$F_c(\mathbf{x}_{\mathrm{obs}}) = \sum_p q_p \int_{[-1.5,1.5]^3} W(\mathbf{s})\, K_{\mathrm{point},c}\big(\mathbf{x}_{\mathrm{obs}} - \mathbf{x}_p - \mathbf{s}\,\mathbf{cell},\ \mathbf{u}\big)\, d\mathbf{s}^3, \tag{1}$$

where $W(\mathbf{s}) = W_x(s_x)W_y(s_y)W_z(s_z)$ is the separable TSC shape (Section 3) and K_{point} is the per-unit-charge field of one uniformly moving charge. Because the integral is linear in the source charge, the sum over particles is exact; the whole design is a fast evaluation of this one integral.

2.1 The moving point kernel

The kernel is the present-time Liénard–Wiechert field of a uniformly moving charge [extraction worker, 2026]. For constant velocity the field points to the instantaneous position. In the lab frame, with $\mathbf{R} = \mathbf{x}_{\mathrm{obs}} - \mathbf{x}_{\mathrm{src}}$,

$$\gamma = \sqrt{1 + |\mathbf{u}|^2}, \qquad \boldsymbol{\beta} = \mathbf{u}/\gamma, \qquad \mathrm{invg}^2 = 1/(1 + |\mathbf{u}|^2) = 1 - \beta^2, \tag{2}$$

$$D = |\mathbf{R}|^2\, \mathrm{invg}^2 + (\mathbf{R}\cdot\boldsymbol{\beta})^2, \tag{3}$$

$$\mathbf{E}_n = \mathrm{invg}^2\, \mathbf{R}/D^{3/2}, \qquad \mathbf{B}_n = \boldsymbol{\beta} \times \mathbf{E}_n. \tag{4}$$

For $\mathbf{u} = \mathbf{0}$ exactly, $\mathbf{E}_n = \mathbf{R}/|\mathbf{R}|^3$ and $\mathbf{B}_n = \mathbf{0}$; no ε is added to any denominator. The kernel is homogeneous of degree -2 in $\mathbf{R}$. A bunch with a velocity spread needs one convolution per velocity class (Section 6.13); the validated scope is a single shared $\mathbf{u}$.

2.2 The unit chain

The reference normalizes the per-unit-charge field as $E_n = 4\pi\varepsilon_0\, E/q$ and $B_n = 4\pi\varepsilon_0 c\, B/q$, so $B_n = \boldsymbol{\beta}\times E_n$ [extraction worker, 2026]. The mesh produces the per-unit-charge field with $\mathbf{R}$ in metres, already summed over the deposited charge q_p in coulombs. The chain to physical SI is

$$E_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0), \quad B_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0 c), \tag{5}$$

with $\varepsilon_0 = 8.8541878128 \times 10^{-12}$ F/m and $c = 299792458$ m/s. The h^3 factor (Section 3) is the Riemann volume of the sampled continuum integral [extraction worker, 2026].

2.3 The TSC shape

W is the quadratic B-spline, with argument in cell units, support $[-1.5, 1.5]$ per axis, and unit integral per axis [extraction worker, 2026]:

$$W(x) = \begin{cases} 3/4 - x^2 & |x| < 1/2, \\ \frac{1}{2}(3/2 - |x|)^2 & 1/2 \le |x| < 3/2, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

3 Method

3.1 The operator interface

`v0.7.0` (commit `a152c518`, tag `25d6ca8b`) is driven by `picBunch -input bunch.h5 -destination <checkpoint> -config params.toml`, with every unit and conversion parameter in the TOML file [placement worker, 2026]. The `[picongpu]` section carries `unit_length`, `unit_e`, `unit_b`, `dt`, `step`, `precision`, `species`; `[domain]` carries `cells`, `cell_size`, `origin`; `[placement]` carries `offset_unit` and the `left`/`front`/`top` fractions; `[accuracy]` carries `tolerance` and the near-shell settings, `near_shell none`/`exact`/`fast` plus `near_shell_radius` and `near_shell_tile`. A missing conversion key causes a clear error at start-up. The input bunch is centred by its charge-weighted centroid and placed at the requested offset from the domain origin; a placement leaving the bunch outside the domain is rejected at start-up. `patch_count` in `[placement]` splits the species into one particle patch per output rank, with cumulative, disjoint `numParticlesOffset` ranges, for multi-rank restarts (Section 7) [placement worker, 2026, mpi worker, 2026a].

The default `precision` is uniform `float32` for every field and particle record, and such a checkpoint restarts under a stock single-precision PIConGPU; `float64` is a uniform opt-in that records a warning attribute because stock PIConGPU cannot load double records [precision worker, 2026]. The `charge` and `mass` records are written rather than omitted, and the species is a single particle patch, `patch` = `"bbox"` or `"domain"` [placement worker, 2026].

The mesh evaluates Eq. (1) by a discrete deposit, a sampled kernel, and a free-space FFT convolution. Let N be the bunch-box nodes per axis, $M = \mathrm{pad} \cdot N$ the padded FFT grid per axis, h the node spacing in cell units (node i sits at ih **cell**), and **cell** $= (dx, dy, dz)$. All precision-critical work is in double [scaling worker, 2026a, extraction worker, 2026].

3.2 Deposit

Each source charge is deposited onto the zero-padded M^3 grid with the TSC shape, at the cell-unit spline argument $W((\mathbf{x}_i - \mathbf{x}_p)/\mathbf{cell})$, which is correct for any h [extraction worker, 2026]. The result $\rho[i] = \sum_p q_p\, W(ih - \mathbf{p}_{\mathrm{cell}})$ lives in the N^3 sub-box of the padded grid, indexed with M-stride; the rest is zero. A fixed reduction order is required for MPI-layout invariance; the single-rank path uses `atomicAdd` [extraction worker, 2026].

3.3 Kernel sampling

The six TSC-convolved kernels are sampled on the same padded M^3 grid, one thread per FFT point, with the centered (negative-wrapped) index $d_a = a$ for $a < M/2$ else $a - M$. For each of the six components c, the kernel is evaluated at the physical separation

$$\mathbf{R} = (d_a\, h + \delta_{c,a})\, \mathbf{cell}, \tag{7}$$

where δ_c is the Yee staggered offset (Section 3.5). The kernel is therefore sampled at the same node spacing h as the deposit, with the shape one grid cell of its own axis wide [worker, 2026a]. Non-finite samples are set to zero to avoid $0 \times \infty$ in the FFT [extraction worker, 2026].

3.4 FFT convolution and normalization

The convolution is a free-space, zero-padded FFT convolution. The vendor plan is built once, unnormalized, and reused for all thirteen transforms: one forward for ρ, six forwards for the kernels, and six inverse transformations [extraction worker, 2026]. The unnormalized round trip is scaled by M^3; the code divides by $h^3 M^3$ in the read-out [extraction worker, 2026].

Padding. The grid must be zero-padded to pad ≥ 2. With pad $= 1$ the cyclic convolution wraps and a 9.275×10^{-3} periodic-image artefact appears for a single stationary particle in a 20^3 box [worker, 2026b]. For a single stationary particle the pad $= 2$ result agrees with the direct non-cyclic sum to 3.811×10^{-16} and the pad $= 3$ result to 1.905×10^{-16} [worker, 2026b]. At pad ≥ 2 the error is at machine precision ($\leq 10^{-15}$) [worker, 2026b]. The artefact grows for beamed sources: at pad $= 1$ the 100-source $\gamma = 10$ cloud shows a 7.5×10^{-1} worst-component error [worker, 2026b].

DC and Nyquist. For the R2C/C2R half-spectra the DC bin and the seven self-conjugate even Nyquist corners must be exactly real; the intermediate Nyquist planes are Hermitian-paired, not real [worker, 2026b]. The R2C/C2R path matches the C2C path to 1.665×10^{-16} over all six components [worker, 2026b].

The h^3 factor. The sampled convolution evaluates the continuum integral of Eq. (1) by a Riemann sum whose nodes are spaced h cells apart, so each sample covers a physical volume h^3 (in cell units); the read-out therefore applies the factor $h_x h_y h_z / (M_x M_y M_z)$ [extraction worker, 2026].

3.5 Read-out

The six field components live at the staggered Yee positions, in cell units [extraction worker, 2026]:

$$\begin{array}{llllll} E_x & (0.5, 0, 0) & E_y & (0, 0.5, 0) & E_z & (0, 0, 0.5) \\ B_x & (0, 0.5, 0.5) & B_y & (0.5, 0, 0.5) & B_z & (0.5, 0.5, 0). \end{array} \tag{8}$$

The kernel is sampled at these offsets, so the convolution is already evaluated on the Yee grid. The read-out compacts the M^3 FFT result to the N^3 bunch box and applies the h^3/M^3 scale and the SI factor (Section 2). No further interpolation is needed [extraction worker, 2026].

3.6 Scale invariance

The kernel is homogeneous of degree -2, the deposit shape width equals the node spacing, and the Yee offsets are in cell units, so every relative metric depends only on r/h, the distance in cell units [worker, 2026a]. At a fixed distance in cells the error is therefore independent of h: refining h with the distance held in cells changes nothing, and the measured relative errors at $h = 1, 1/2, 1/4, 1/8$ are identical to machine precision for $\gamma \in \{1, 10, 100\}$, aspect 1 and 4 [worker, 2026a]. At a fixed physical distance r, refining h increases r/h and does reduce the error, at the cost of M^3 FFT work in $1/h$. The controlling length for a beamed source is the beaming cone width d/γ: where that width is sub-cell, the error is set by the cone, not by the mesh, and refinement stops paying off (Section 6.11) [worker, 2026c].

3.7 The near shell and the fast evaluator

The default mesh is the whole field. `v0.7.0` carries an opt-in near shell (`[accuracy]` in the TOML config, Section 3.1), with a radius and a tile [near-shell worker, 2026, far-tail worker, 2026, fast-near worker, 2026]. It computes

$$F = F_{\mathrm{mesh}}(\mathrm{all}) - F_{\mathrm{mesh}}(\mathrm{near}) + F_{\mathrm{near}}(\mathrm{near}), \tag{9}$$

where the subtraction uses a per-tile local zero-padded FFT of the near sources (padding 2) and the near term is evaluated per (source, node, component) by one of two backends selected by `near_shell`: `exact`, a vendored adaptive FP64 TSC-convolution quadrature [near-shell worker, 2026], or `fast`, the closed-form evaluator below. The default remains `none`; the radius is in mesh nodes of the compute grid, and the envelope it must cover depends on h (Section 6.5). The fast backend is the recommended one: its measured accuracy and cost are in Sections 6 and 6.8 [fast-near worker, 2026].

The fast evaluator splits the in-support integral of Eq. (1) into a singular part and a smooth remainder. The singular part is the eight-term `asinh` box of the kernel's divergence, O(1) per pair, built from the double integral

$$\begin{aligned} T(A,B,u) &= \int_0^A \int_0^B \frac{\mathrm{d}X\,\mathrm{d}Y}{\sqrt{u^2+X^2+Y^2}} \\ &= \begin{cases} A\,\mathrm{asinh}(B/A) + B\,\mathrm{asinh}(A/B), & |u| < 10^{-300}, \\ A\,\mathrm{asinh}\big(B/\sqrt{u^2+A^2}\big) + B\,\mathrm{asinh}\big(A/\sqrt{u^2+B^2}\big) & \\ \quad -u\,\mathrm{atan}\big(AB/(u\sqrt{A^2+u^2+B^2})\big), & \text{otherwise.} \end{cases} \end{aligned} \tag{10}$$

with $A, B > 0$ (each box face has 8 corners, 24 such calls per component, so the eight-term box is O(1) per pair). The form is even in u, so the $|u| \to 0$ branch must fire on $|u|$, not u (a negative u is a field point beyond a box face) [fast-near worker, 2026]. Verified against a converged midpoint trapezoid reference (-2.6348 converged; the naive trapezoid -2.6333 is 1.5×10^{-3} low, a trapezoid singularity artefact [fast-near worker, 2026]). The primitive is implemented in the fast evaluator header `include/picBunch/detail/FastNear.hpp` and the device reference `tools/closed-form/gpu_extract.cu` at picongpu `b3699df` [fast-near worker, 2026, closed-form worker, 2026]. The remainder is a plain tensor Gauss–Legendre quadrature on the subtracted integrand $(S - W^*)K$ over the TSC knot panels, each axis split at the TSC knots and at the anchor point s^*; the default order is 5, 8000 nodes per in-support pair [fast-near worker, 2026, evaluator worker, 2026]. For a moving bunch the box is γ-stretched for the axial direction, and the oblique direction uses a rest-frame octant quadrature of the singular box at order 8, 4096 nodes, plus the same remainder. Host and device use the same algebra [fast-near worker, 2026].

The assembly is exact. Single-source tests at a tile centre and a tile boundary give an assembly residual of 2×10^{-12} to 1.5×10^{-11} SI (4×10^{-16} relative), flat with distance. A local FFT of the near set against the global FFT agrees at 10^{-15} for margins 0, 2, 4 and a 100-particle cloud, and boundary accounting shows the same near set feeding the exact sum and the subtracted mesh (field identity 3.6×10^{-13}). A direct reproduction with 100 particles at radius 3 gives an assembled field exact to 4.6×10^{-13} over the whole grid, while the plain mesh on the same plane is wrong by 5.6×10^{-1} [worker, 2026d].

The fast backend changes no far number: the mesh-only field from the fast and the exact backends is bitwise identical (same md5), and the total field outside the near tiles is bitwise equal, so the tile subtraction stays intact [fast-near worker, 2026]. The seam jump excess at the fast/exact crossing, on a 16^3 grid, $R = 3$ cells, tile 4, cells 10^{-7} m, is 2.03×10^{-10} per component for a stationary single particle (2.02×10^{-10} on $E_x/E_y/E_z$, zero on B) and 2.8×10^{-10} for the 64-source cloud [fast-near worker, 2026, release worker, 2026].

3.8 MPI parallelism

`v0.7.0` writes a global output domain larger than the compute box [mpi worker, 2026b, far-tail worker, 2026], and computes the field in parallel across ranks [rank-parallel worker, 2026]. The written extent is set by `-global-grid` and `-global-origin`. The global array is never allocated: each rank owns a contiguous z-slab split into tiles of about 128 MiB, zero-filled, then filled with the compute-box overlap, the rest exact 0.0. All ranks open one openPMD series with `MPI_COMM_WORLD` and write disjoint

Table 1: Far-tail switch validation, 20-cell 4096-electron bunch, $R = 16.85$ cells, tolerance level 45.72 V/m (10^{-3} of the peak 45722.5 V/m). Worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum over 13 directions [far-tail worker, 2026]. The switch at $6R$ is the one shipped: the error is 1.87×10^{-3} of the tolerance level.

switch radius	worst err (V/m)	/ tolerance level
$2R$ (33.7 cells)	4.80×10^{1}	1.05
$4R$ (67.4 cells)	8.77×10^{-1}	1.92×10^{-2}
$6R$ (101.1 cells)	8.53×10^{-2}	1.87×10^{-3}
$8R$ (134.8 cells)	1.66×10^{-2}	3.63×10^{-4}

`storeChunk` regions with collective flushes, so the output is a single dataset, not per-rank files. A `-deterministic-deposit` flag gives a serial host deposit for bitwise comparisons [mpi worker, 2026b].

The compute is parallel in two ways, neither of which is a domain-decomposed FFT [rank-parallel worker, 2026]. The mesh runs six independent component convolutions: every rank deposits and transforms the same ρ spectrum (replicated), rank r evaluates the components c with $c \bmod \text{size} = r$, and one `MPI_Allreduce` sums the six arrays. The near shell partitions the output tiles by linear tile index; each rank builds its own near-source list and local subtraction FFT for its tiles and writes only those corrections, and one `MPI_Allreduce` sums the correction arrays. The bunch is read once on rank 0 and broadcast, not redistributed. Because the mesh components are disjoint and each near correction element is owned by exactly one rank, the all-reduce order cannot change a value: the outputs are bitwise identical at 1, 2 and 4 ranks (Section 4.1) [rank-parallel worker, 2026, mpi worker, 2026a].

3.9 The far-tail margin

For a global domain that extends beyond the compute box, the zero fill must not start until the field is below tolerance. `v0.7.0` places that boundary at the smallest centre distance d where

$$E_{\text{tail}}(d) = \gamma_{\max}\, k \left(\frac{|Q|}{d^2} + \frac{2|\mathbf{p}|}{d^3} + \frac{3.5\,\|D\|_2}{d^4} \right) \le \text{tolerance} \cdot E_{\text{peak}}, \tag{11}$$

with Q the total charge, $\mathbf{p}$ the dipole, $D_{ij} = \sum q\,(3x_i x_j - |\mathbf{x}|^2 \delta_{ij})$ the traceless quadrupole from the actual particle positions, $\|D\|_2$ its largest absolute eigenvalue, and $\gamma_{\max} = \max \sqrt{1 + |\mathbf{u}|^2}$ [far-tail worker, 2026]. Beyond a measured switch radius the writer instead evaluates the multipole field directly (`FarTail.hpp`), so the shipped file has no interior zero fill [far-tail worker, 2026]. The margin is reported at run time as a bound in V/m.

For the documented 20-cell, 4096-electron bunch (source radius $R = 16.85$ cells, peak 45722.5 V/m, tolerance level 45.72 V/m), the quadrupole margin is 350 cells, which exceeds the 2^{28} padded-node cap, so the code falls back to the measured far-tail switch. Table 1 gives the switch error: the worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum in 13 directions [far-tail worker, 2026].

A shipped 512^3 output sampled outside the compute box confirms the switch in practice: 36 nodes including corners, worst $|\mathbf{E}_{\text{file}} - \mathbf{E}_{\text{direct}}| = 2.8 \times 10^{-3}$ V/m against the 45.7 V/m tolerance level (6.1×10^{-5} of it), and every sample nonzero, i.e. no interior zero fill [far-tail worker, 2026].

4 Measured scaling

All timings are on one host, `rosi_v100` (node `gv002`), exclusive Slurm allocation, pinned to GPU 0 [scaling worker, 2026a]. Build: CUDA 12.6, `PICBUNCH_ARCH=70`, Release, padding 2, double [scaling worker, 2026a]. Each configuration ran three timed repeats after one untimed warm-up plus one device-level warm-up; stage values are medians, the total is min/median/max [scaling worker, 2026a]. The four $\gamma = 10$, $N = 2 \times 10^6$ repeats give total medians 60.35 to 64.76 ms, about 7% scatter; the 64.81 ms row is the $\gamma = 100$ run [scaling worker, 2026a]. These compute-only totals exclude the checkpoint writer; the end-to-end totals with the writer are in Table 3, a different measurement path.

Reproducibility of Table 2. The matrix is the `picBunchBench` sweep of picBunch branch experim ental/bench at 3d05752a493e12f3a0a03c6754f92f202a70a482 (picBunchBench--sweep--repeat3--csvmatr

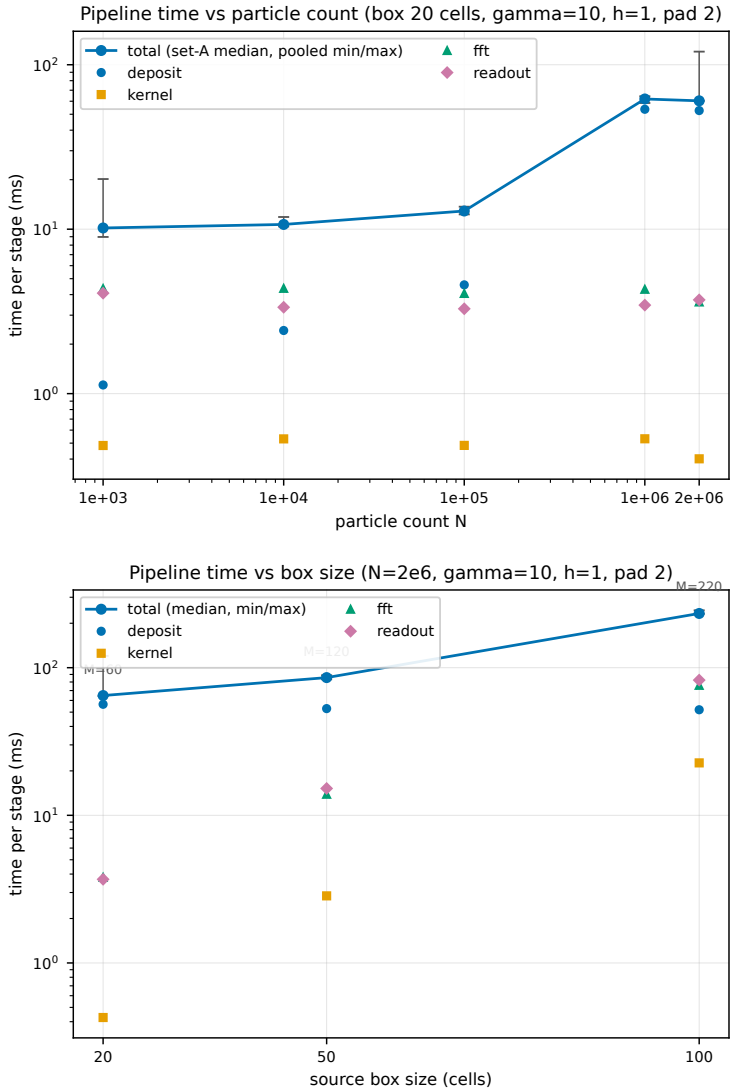


Figure 1: Pipeline time vs particle count (top) and vs source-box size, padded array $M = 60/120/220$ cells (bottom); log-log; total, blue, with the per-stage medians (deposit, kernel, FFT, read-out) side by side, not stacked; $\gamma = 10$, $h = 1$, double; `rosi_v100` (V100-SXM2-32GB), CUDA 12.6, Release [scaling worker, 2026a]. The deposit dominates at $N = 2{\cdot}10^6$, 57 of the 64.8 ms total, while the total grows 64.8 ms to 233 ms from $M = 60$ to $M = 220$.

ix.csv); the bench source and `docs/benchmarking.md` are on that branch [scaling worker, 2026a]. The input bunches are not checked into the repository; the bench source at `3d05752a` is the generator [scaling worker, 2026a]. The fast GPU kernel tools/closed-form/gpu_extract.cu lives in picongpu at `b3699df` [fast-near worker, 2026]. The H100 runs used CUDA 13.4 [fast-near worker, 2026, release worker, 2026].

Correctness anchor. Every timing run is anchored by the Coulomb limit ($\mathbf{u} = 0$): one particle at a node, far nodes at $d = 8, 16, 24, 32$ along all six axis directions. The mesh field against the direct TSC stencil sum of the analytic Coulomb kernel has worst relative error 4.88×10^{-14} (max absolute 2.28×10^{-25}), passing the 10^{-12} threshold [scaling worker, 2026a,b]. This confirms the zero-padded FFT convolution reproduces the TSC-convolved Coulomb field to roundoff.

Fig. 1 sweeps one axis at a time: the top panel fixes the 20-cell box and varies N, the bottom panel fixes $N = 2{\cdot}10^6$ and varies the box. The scaling is in two regimes, with the crossover set by the box, not the particle count [scaling worker, 2026a]. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows while the FFT, kernel, and read-out stay under 10 ms at the 20-cell box; the total is flat across $\gamma = 1, 10, 100$ (64.9/64.4/64.8 ms) [scaling worker, 2026a]. At large boxes the kernel and FFT scale with M^3: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs

Table 2: Per-stage medians (ms) and device allocation (MiB), padding 2, double, `rosi_v100` GPU 0 [scaling worker, 2026a]. "box" is the source-box size in cells; M is the padded FFT array per axis. Stage columns are medians over the three timed repeats.

N	box	γ	h	n	M	total	deposit	kernel	FFT	read-out	dev
10^3	20	10	1	30	60	10.17	1.13	0.48	4.36	4.08	8.4
10^4	20	10	1	30	60	10.69	2.42	0.53	4.37	3.35	8.7
10^5	20	10	1	30	60	12.90	4.59	0.48	4.07	3.28	11.4
10^6	20	10	1	30	60	61.92	53.61	0.53	4.31	3.45	38.9
$2{\cdot}10^6$	20	10	1	30	60	64.76	56.51	0.43	3.80	3.69	69.4
$2{\cdot}10^6$	50	10	1	60	120	85.67	52.82	2.85	13.89	15.20	127.4
$2{\cdot}10^6$	100	10	1	110	220	232.97	51.89	22.66	75.97	82.40	468.7
$2{\cdot}10^6$	20	10	1/2	50	100	69.27	52.38	1.29	7.43	8.16	99.5
$2{\cdot}10^6$	20	10	1/4	90	180	151.51	50.94	11.76	43.38	44.58	284.5

Table 3: End-to-end pipeline time including the checkpoint writer and the openPMD reader, medians over 3 repeats, `rosi_v100` GPU 0, 20-cell box, $\gamma = 10$, shipped CLI `-tolerance` 10^{-3} [worker, 2026e]. A different measurement path from the compute-only Table 2: the totals include the writer (the dominant added term); the per-stage split is deposit/kernel/FFT/readout/writer. The $N = 10^6$ total is 47.15 ms here versus 61.92 ms in the compute-only campaign.

N	total	deposit	kernel	FFT	readout	write
10^4	22.00	2.70	0.53	4.53	3.06	11.29
10^5	24.87	5.82	0.56	4.31	3.80	10.65
10^6	47.15	31.56	0.46	4.20	3.95	7.54

1.07× and 2.35× because the deposit stays constant [scaling worker, 2026a]. The per-h totals are 64.76, 69.27, and 151.51 ms for $h = 1$, 1/2, 1/4 (padded array $M = 60/100/180$, 2×10^6 particles, 20-cell box) [scaling worker, 2026a]. On the `H100` (`jureca_h100`, `sm_90`), the same 2×10^6, 20-cell, $h = 1$ mesh is 0.052 s compute (deposit 34.4 ms, kernel 1.08 ms, FFT 7.44 ms, read-out 9.02 ms; medians of three), and 0.755 s CLI wall including the particle read and the HDF5 write [release worker, 2026].

No configuration came near the 32 GB device limit; the largest exact device allocation was 468.7 MiB [scaling worker, 2026a]. Memory scales with the padded bunch box, not the global field.

Table 3 is the same pipeline with the writer in the loop. The deposit is still linear in N (2.70 → 31.56 ms from 10^4 to 10^6), and the FFT is still box-bound and flat. The writer is the dominant added term: 11.29 ms at 10^4 and 7.54 ms at 10^6. The $N = 10^6$ total is 47.15 ms with the writer, versus 61.92 ms in the compute-only Table 2; the two campaigns differ in measurement path and timing boundary, so neither is the other's total plus the writer [worker, 2026e, scaling worker, 2026a].

4.1 MPI rank scaling

The rank-parallel compute (Section 3.8) and the chunked writer were measured on an idle `rosi_v100` window, medians of three runs [mpi worker, 2026a]. The node was checked idle before the block (no CPU worker, 1-min load below 2), but each rank block starts from the preceding measurement, so the 1-min load rises within the block: the near-shell block starts at 1.9, 8.4 and 11.8 for 1, 2 and 4 ranks (mesh block 1.4–1.5, write block 1.1–1.6); no foreign process was active during the window [mpi worker, 2026a]. Correctness was verified at 1, 2 and 4 ranks on a 128-particle, 32^3, cell 8×10^{-7} m, tolerance 10^{-3}, near-shell $R = 1.5$, tile 8 problem: the mesh and the near shell are bitwise identical at 1-vs-2 and 1-vs-4, all six components, $\max_{\text{abs diff}} = 0.0$ [mpi worker, 2026a, rank-parallel worker, 2026].

Table 4 gives the speedup for the three parallel pieces. The mesh is component-parallel: the replicated deposit (307–332 ms, flat in rank count) and the replicated source-spectrum FFT cap it, so the speedup plateaus at 1.20× at both 2 and 4 ranks. The near shell is tile-parallel: the static round-robin tile split plus the all-reduce of the full correction array plateaus it at 1.34× at 4 ranks. The writer is a collective HDF5 write: it improves to 1.71× at 2 ranks and falls to 0.48× at 4 ranks, where the collective flush and metadata traffic contend on the shared filesystem; this is an I/O limit, not a compute limit [mpi worker, 2026a].

Table 4: Rank scaling on `rosi_v100`, idle window, medians of three [mpi worker, 2026a]. Mesh: $N = 2\times10^6$, 112^3, component-parallel (replicated deposit and source FFT). Near shell: $N = 128$, 32^3, $R = 1.5$, tile 8, tile-parallel. Write: $N = 2 \times 10^6$, 256^3 global, 0.77 GB, collective HDF5. Mesh and near shell are bitwise identical at 1/2/4 ranks.

	near shell			mesh		write	
ranks	wall (s)	speedup	eff.	wall (ms)	speedup	wall (s)	speedup
1	36.61	1.00	100%	522.2	1.00	7.41	1.00
2	29.60	1.24	62%	436.3	1.20	4.34	1.71
4	27.29	1.34	34%	433.6	1.20	15.49	0.48

5 Measured error structure

The measured error depends on the observation point, the direction relative to the beam, and γ. We report the regimes in which the error is largest, together with the configuration and reference for each number.

The fast near backend, against the independent continuum, is 7.99×10^{-4} stationary interior, 1.96×10^{-5} face, and 1.57×10^{-7} outside; 1.69×10^{-4} moving axial interior, 1.11×10^{-4} face, 2.12×10^{-6} outside; and 8.15×10^{-2} moving oblique interior, the measured limit of the fast backend (Section 6.6) [fast-near worker, 2026, release worker, 2026]. The oblique value is the one region where the fast backend is above 10^{-3} at its default order.

The mesh alone, without the near shell, reproduces the field to 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$ where the slices cut the bunch core (Section 6.5); these planes cut the sources, the region where the single-mesh spectral convolution has no near-field treatment. With the near shell enabled at the measured radii ($R = 3$ for $\gamma = 1$, $R = 4$ for $\gamma = 10$ in mesh nodes of the $h = 1$ slice grid), the assembly residual is at roundoff and the measured plane error is 1.6×10^{-3}–1.1×10^{-1} for $\gamma = 1$ (the high end is the z-y plane E_z, 0.1067) and 3.2×10^{-2}–5.7×10^{-1} for $\gamma = 10$ (Section 6.7); the residual is the mesh contribution outside the shell radius.

In the perpendicular direction at $\gamma = 10$, aspect 1, the measured $M-T$ error at $d = 48$ is 2.6×10^{-4}, the $M-P$ error is 1.5×10^{-2}, and the TSC smoothing $T-P$ is 1.48×10^{-2}, of which the $M-P$ error is the TSC smoothing, not the deposit (Table 5) [worker, 2026a, Orchestrator, 2026a,b]. At $\gamma = 100$, aspect 1, the perpendicular $M-T$ error at $d = 48$ is 3.75×10^{-1} and does not decrease with further increase in d [worker, 2026a, Orchestrator, 2026c]. At aspect 4, $\gamma = 1$, the perpendicular $M-T$ error at $d = 48$ is 2.4×10^{-3} [worker, 2026a]. The axial direction is below 3×10^{-4} at $d = 6$ for every γ [worker, 2026a].

The γ-landscape measurement at $d = 8$ cells, $h = 1$, shows the error 4.4×10^{-2}, 1.05, 5.47 for $\gamma = 10, 50, 100$. The h-refinement exponent p (error $\sim h^p$, fitted from $h = 1$ and $h = 1/4$) falls from 1.85 to about 1.0 as γ rises; the controlling length is the beaming cone width d/γ, and at $\gamma \geq 50$ it is sub-cell (Section 6.11) [worker, 2026c]. The detailed discussion is in Section 8.

6 Accuracy

Each error figure names its reference (M mesh, T continuum, P point). The three references stay apart throughout: the mesh M, the TSC-convolved continuum T (Eq. (1), the quantity the mesh is built to compute), and the point field P (the un-smoothed Liénard–Wiechert field). M against T is the discrete deposit and convolution error. T against P is the TSC smoothing, an $O(1)$ effect at $\gamma h \gg 1$. M against P is the total.

6.1 The reference and the metrics

The arbiter is an independent TSC-convolved continuum, a tensor Gauss–Legendre quadrature with a TSC-knot split and a singularity-cancelling power map [pr24w53 ref, 2026]. Its quadrature error is below the 3×10^{-4} gate in every configuration measured here [worker, 2026a].

Three metrics are reported and are not interchangeable, and each claim in this section names which one supports it [worker, 2026f].

Floored. $|\text{ref} - \text{mesh}| / \max(|\text{ref}|, |\text{mesh}|)$, over points where $|\text{ref}| \geq 10^{-3}$ of the local (six-component) field peak. This is the relative per-component error of record for the electric field.

Table 5: L2-pooled relative error at $d = 48$ cells, $h = 1$, perpendicular direction (L2 over the six components; single-component worst: $\gamma = 1$, aspect 4, $M-T$ 2.65×10^{-5}; $\gamma = 10$, aspect 1, 1.4×10^{-3}) [worker, 2026a]. $M-T$: mesh vs TSC-convolved continuum; $M-P$: mesh vs point field; $T-P$: TSC smoothing, mesh-independent. At $\gamma = 10$, aspect 1 the $M-P$ vs $M-T$ gap is the smoothing (L2 ratio 58×), not the mesh.

γ	aspect	$M-T$	$M-P$	$T-P$
1	1	1.38×10^{-7}	9.89×10^{-8}	3.96×10^{-8}
1	4	2.65×10^{-5}	2.39×10^{-3}	2.40×10^{-3}
10	1	2.6×10^{-4}	1.5×10^{-2}	1.48×10^{-2}
10	4	3.33×10^{-2}	1.19×10^{-1}	1.44×10^{-1}
100	1	3.75×10^{-1}	2.20×10^{-1}	4.40×10^{-1}
100	4	2.78×10^{0}	2.49×10^{-1}	8.32×10^{-1}

Table 6: Worst per-component relative error (over x, y, $|\text{ref}| \geq 10^{-3}$ of the local field) of the shipped CLI against the TSC-convolved continuum, at $d = 16$ and $d = 32$ cells; rosi_v100, `indfield.py` order 12 refine 3 [worker, 2026e]. "s/c" = single / cloud. Measured values at $d = 48$: $\gamma = 10$, 1.8×10^{-3} ($d = 32$: 6.5×10^{-3}); $\gamma = 1$ is below 3×10^{-4} from $d = 12$ (aspect 1) and $d = 32$ (aspect 4).

γ, aspect	s/c	d=16	d=32
1, 1	3.5e−5 / 1.0e−4	2.2e−6 / 2.0e−5	
1, 4	2.8e−3 / 2.1e−3	2.0e−4 / 2.6e−4	
10, 1	5.6e−2 / 5.3e−2	6.3e−3 / 6.5e−3	
10, 4	5.5e−2 / 5.0e−2	6.3e−3 / 6.4e−3	

Unfloored. The same ratio at every point. It is the only one that reaches the magnetic field, because the floor excludes it (below).

Absolute. $|\text{ref} - \text{mesh}| / \max_{\text{point}} |\text{ref}_c|$, the peak-normalized error; it supports whole-field statements.

The judge metric $|\text{ref} - \text{mesh}|/(1 + \max(|\text{ref}|, \text{mesh}|))$ is absolute-dominated: for fields of order 10^{-9} V/m the +1 denominator swamps the error, so it reports a crossover at $d = 2$ for every case and does not discriminate [worker, 2026e].

The floor hides the magnetic field. For $\gamma = 10$ every B point is excluded by the floor: $|\mathbf{B}|$ is about 10^{-10} of the E-dominated local field, far below the 10^{-3} threshold, so the floored metric reports no magnetic accuracy at all. The unfloored B relative error is 9.7×10^{-4} at $d = 32$ (single particle, aspect 1), against a B peak that the floored column never sees [worker, 2026f]. The floored column of Table 6 is therefore electric-only at $\gamma = 10$.

6.2 Reference-separated error at 48 cells

Table 5 shows two things. At $\gamma = 10$, aspect 1, the mesh against the point field ($M-P = 1.5 \times 10^{-2}$) is about 58× the mesh against the TSC-convolved continuum ($M-T = 2.6 \times 10^{-4}$, an L2 ratio); the difference is the TSC smoothing, which $M-P$ inherits and $M-T$ does not. A mesh-versus-point comparison measures the shape the deposit applies, not the discretization error. Second, the $M-T$ error is not converged perpendicular to the beam at $\gamma h \gg 1$: at aspect 1 it is acceptable only for $\gamma = 10$ and not for $\gamma = 100$; at aspect 4 it is not converged at any γ. Only the axial direction is clean at every γ [worker, 2026a].

6.3 The independent validation

A second, independent pass measured the shipped CLI end to end: `picBunch` at `bfaa1e26`, rosi_v100, `indfield.py` order 12 refine 3, $d = 2$ to 32 cells, $\gamma \in \{1, 10\}$, aspect 1 and 4, single particle and three-particle cloud [worker, 2026e].

Table 6 confirms the reference-separated picture independently. The measured value is 1.1×10^{-4} at $d = 12$ for $\gamma = 1$, aspect 1, and 2.6×10^{-4} at $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$, aspect 1 the relative error is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$. These are floored electric values; the unfloored magnetic error at $\gamma = 10$ is 9.7×10^{-4} at $d = 32$. The window starts at $d = 2$, so the TSC-support

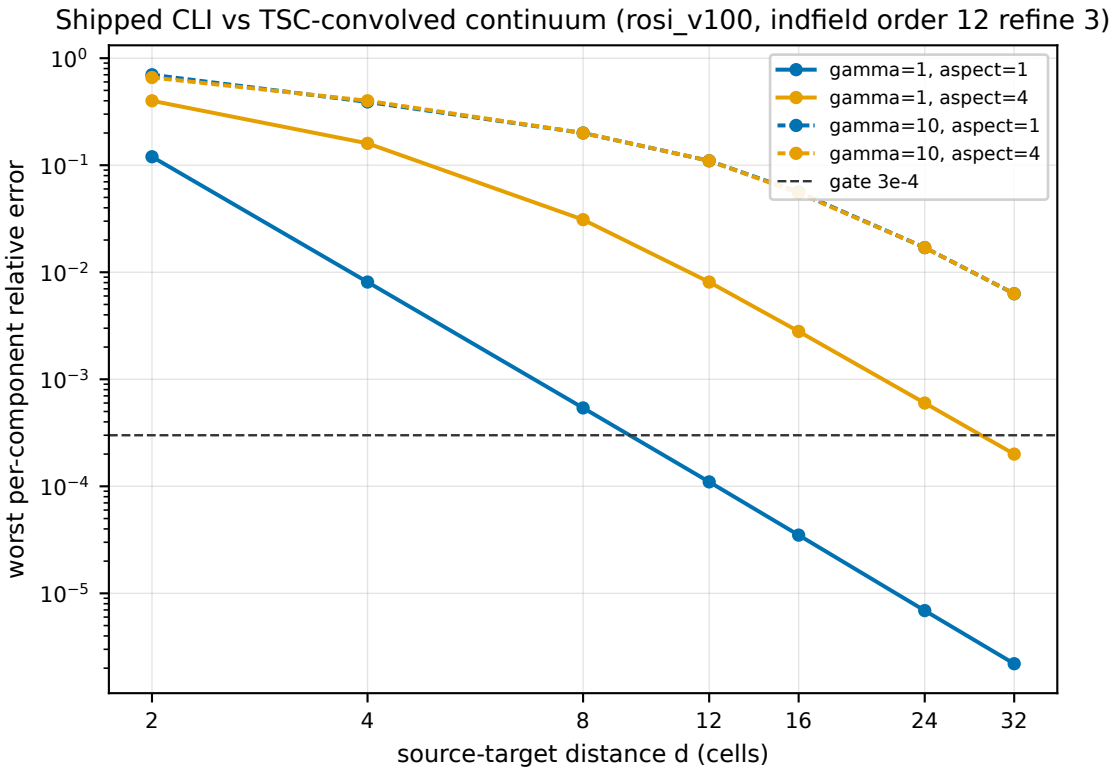


Figure 2: Worst per-component relative error vs source-target distance d in cells, single-particle series, floored electric series, solid $\gamma = 1$ and dashed $\gamma = 10$, blue aspect 1 and orange aspect 4; `rosi_v100`, `indfield.py` order 12 refine 3 [worker, 2026e], reference T. The error falls from 1.2×10^{-1} at $d = 2$ to 2.2×10^{-6} at $d = 32$ for $\gamma = 1$, aspect 1, and is 6.3×10^{-3} at $d = 32$ for $\gamma = 10$, aspect 1.

interior ($|s| < 1.5$ cells) is not covered by these sweeps; it is what the near shell measures (Sections 3.7, 6.8) [worker, 2026e,f,d].

Fig. 2 plots the mesh against the TSC-convolved continuum; the three-particle cloud series is in Table 6. The series is floored, so the magnetic field, which the floor excludes at $\gamma = 10$ (Section 6), is not in the figure. The $\gamma = 10$ point at $d = 48$ (1.8×10^{-3}) is in Table 6 but outside the plotted window. The point-field column of Table 5 ($M-P$) is the mesh against the un-smoothed field, which carries the TSC smoothing in addition to the deposit error and never agrees with $M-T$ for $\gamma h \gg 1$.

6.4 Crossover radii

At $\gamma = 1$, aspect 1, the mesh is within 3×10^{-4} of both references from $d = 8$; the two are interchangeable there. From $\gamma = 10$ up, against the point field the measured perpendicular error at $d = 48$ is 6.5×10^{-3} ($\gamma = 10$, aspect 1); against the TSC-convolved continuum the perpendicular crossover is $d = 48$ ($\gamma = 10$, aspect 1), and for the remaining configurations the measured value at $d = 48$ exceeds the gate. The axial direction is 6 against the continuum at every γ [worker, 2026a].

6.5 Mid-bunch slices: field and error side by side

The distance sweeps above sample the far field. To measure the mesh in the bunch core, we slice the mesh through the core and compare, pixel by pixel, with the independent exact TSC driver [beam-direction worker, 2026, slice worker, 2026a]. The case is $N = 1000$ in a $32 \times 32 \times 16$ box, cell 2×10^{-7} m: a $\gamma = 10$ bunch with momentum along $+y$ ($u_y = 9.95$) and the stationary $\gamma = 1$ bunch. In every panel the horizontal axis is y, increasing left to right, so the bunch travels to the right. The vertical axis is x in the x-y cut and z in the z-y cut. Each figure draws the x-y plane at $z = 1.60\,\mu$m (upper image) and the z-y plane at $x = 3.20\,\mu$m (lower image), both through the bunch centre. Every image has the field row on top and the error row underneath. The electric figures have three columns (E_x, E_y, E_z); pixels below 10^{-3} of the plane peak are masked white, and the error row uses a log scale normalised to the plane

Table 7: Crossover radius in cells: first d at which every larger d is at or below the gate, $h = 1$, L2 over the six components, per axis direction, worst over the three positive axis directions (>48: beyond the measured window) [worker, 2026a]. T: TSC-convolved continuum; P: point field. Oblique directions are $\sim 1.5\times$ larger. The $\gamma = 1$ aspect 1 row is $d = 8$ against both references here, versus the $d = 12$ of the shipped-CLI series in Table 6: the two use different component sets (six axis directions, L2-pooled, versus per-component over x, y) and different implementations [worker, 2026a,e].

	3×10^{-4}		10^{-3}			
γ	aspect	ref	axial z	perp x	axial z	perp x
1	1	T	8	8	6	6
	1	P	8	8	6	6
1	4	T	6	24	6	16
	4	P	48	> 48	32	> 48
10	1	T	6	48	6	48
	1	P	> 48	> 48	32	> 48
10	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48
100	1	T	6	> 48	6	> 48
	1	P	> 48	> 48	32	> 48
100	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48

peak. Fig. 3 is the $\gamma = 10$ electric case, and Fig. 4 its magnetic companion (B_x, B_z; $B_y \equiv 0$ for motion along $+y$ and is omitted). Fig. 5 is the stationary bunch: $B \equiv 0$, so no magnetic companion exists.

Shell radius unit. These slices run the default mesh with no near shell. The near-shell envelope measured in the crossover work, "3 to 4 PIC cells," assumed a mesh spacing $h = 1/4$ of the PIC cell, so $R = 3$ and $R = 4$ are 12 and 16 mesh nodes [near-shell worker, 2026, slice worker, 2026b]. On this grid the node spacing is $h = 1$ PIC cell (2×10^{-7} m), so the same $R = 3/4$ is $3/4$ mesh nodes, four times smaller in physical units than the envelope configuration. At $R = 3$ on an $h = 1$ grid the shell covers 3 mesh nodes; the TSC support extends ± 1.5 cells from each source, so the shell covers a fraction of the full support. The measured plane errors at this radius are reported in Section 6.7.

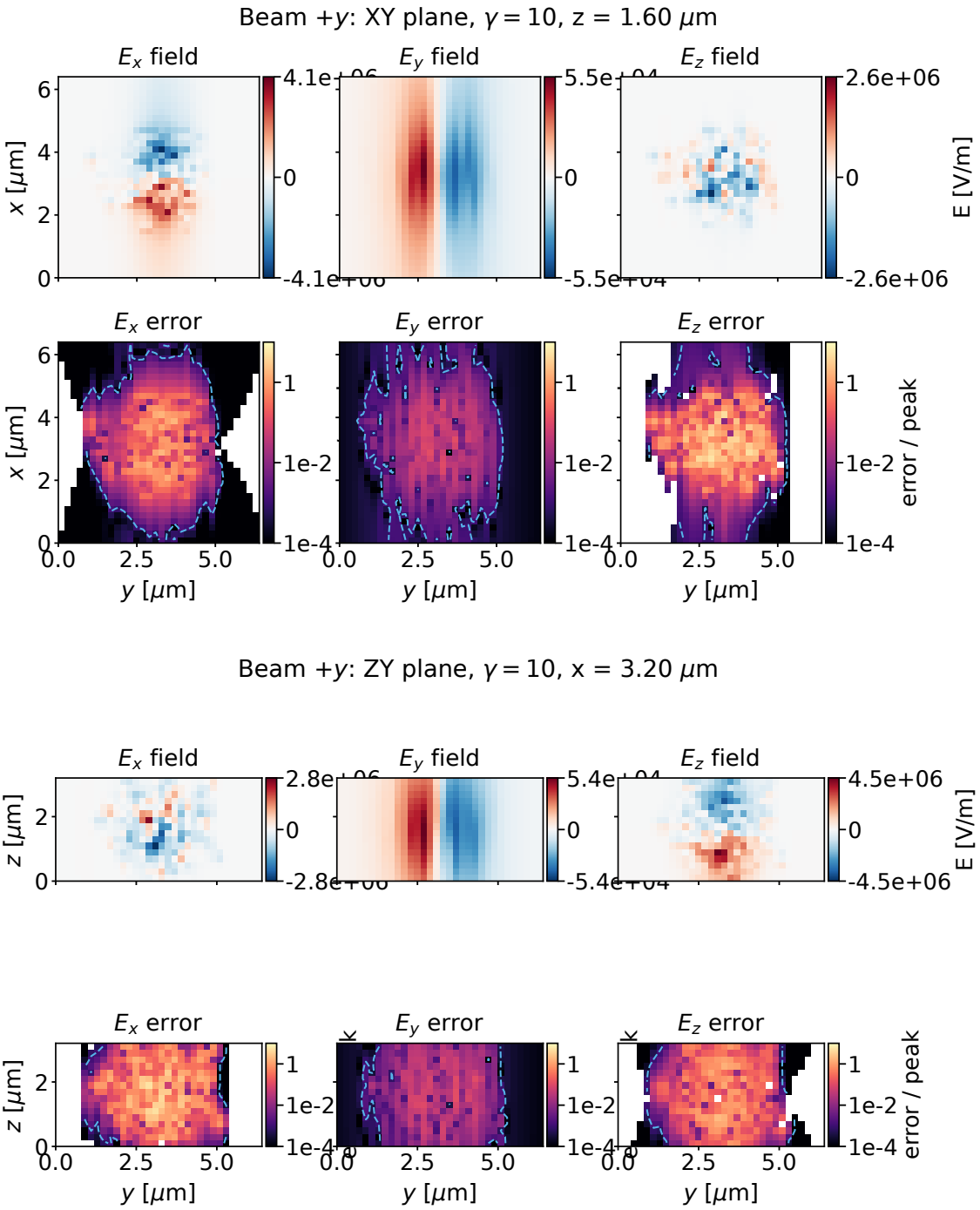


Figure 3: Mid-bunch electric slices, $\gamma = 10$ with momentum along $+y$, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, $h = 1$ PIC cell, no near shell [beam-direction worker, 2026]. The beam axis is y and runs left to right in every panel, so the bunch moves rightwards; the vertical axis is x in the x-y cut and z in the z-y cut, both in µm. Upper image, x-y plane at $z = 1.60$ µm; lower image, z-y plane at $x = 3.20$ µm. In each image the top row is the mesh field and the bottom row the relative error against the exact TSC driver (reference T), log scale; columns are E_x, E_y, E_z. The error in the bunch core is 0.12 to 5.33 per component (Table 8).

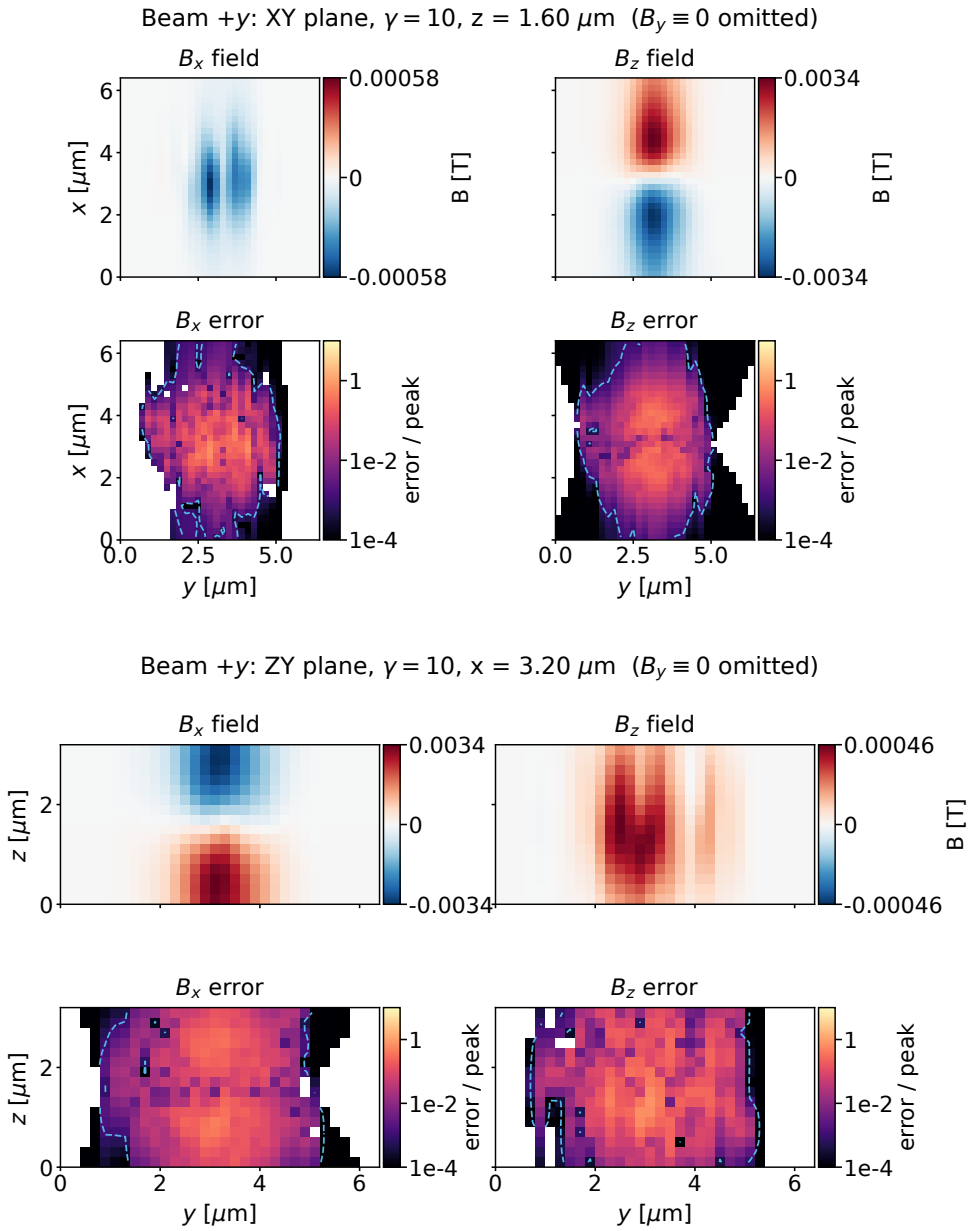


Figure 4: Magnetic companion to Fig. 3, $\gamma = 10$ with the same $+y$ beam axis and error convention. Two columns, B_x and B_z, field row on top and error row below. $B_y \equiv 0$ for motion along $+y$ (measured max $|B_y| = 0$, ratio 0 to the largest magnetic component) and is therefore omitted; the beam moves left to right along y.

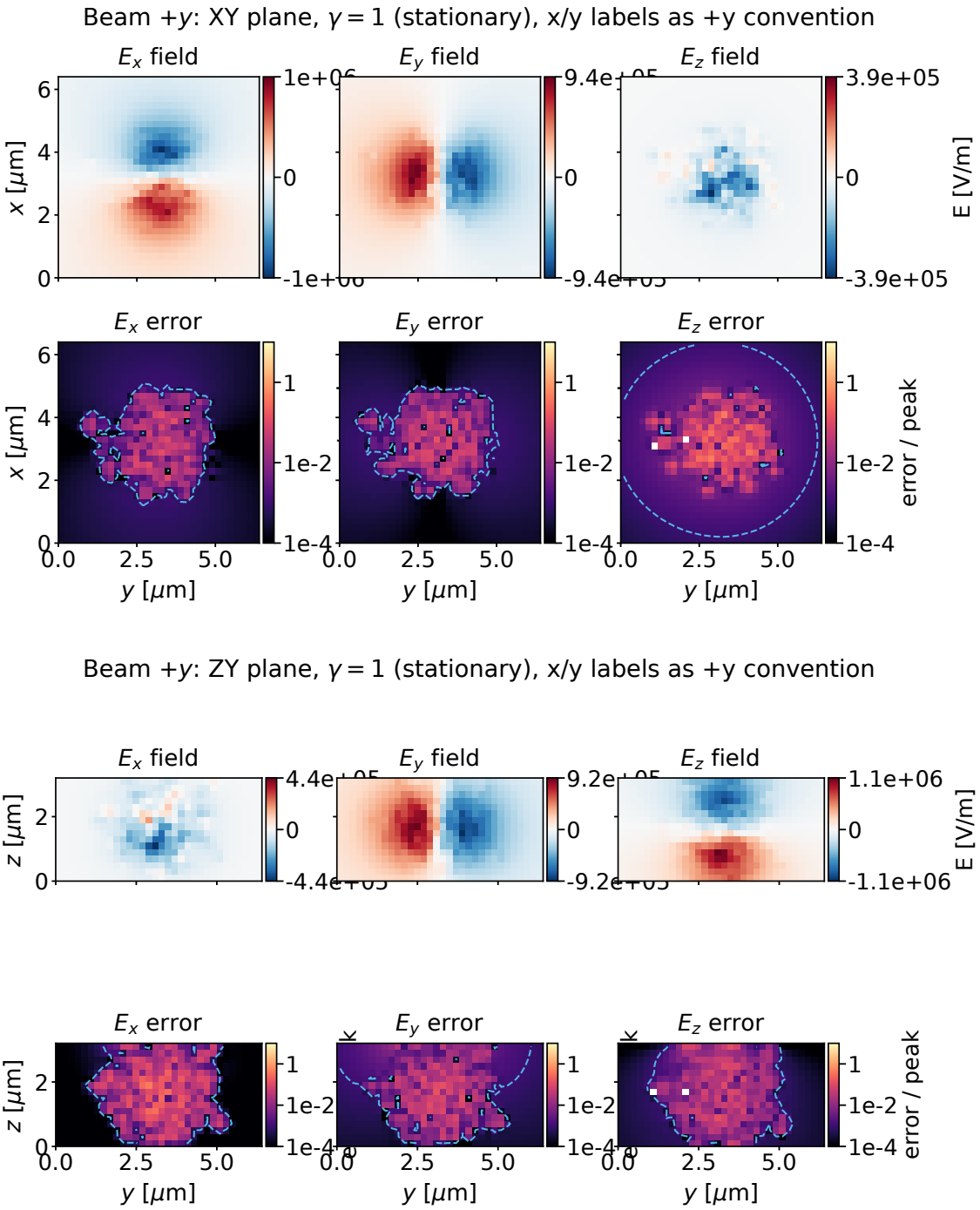


Figure 5: Mid-bunch electric slices, stationary bunch ($\gamma = 1$), $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, $h = 1$ PIC cell, no near shell, same planes, $+y$ beam-axis convention and layout as Fig. 3 [slice worker, 2026a]; y increases left to right and the vertical axis is x (x-y) or z (z-y). The three electric components are drawn, with the field row on top and the error row below. $B \equiv 0$, so no magnetic companion exists. The error in the bunch core is 0.12 to 0.415 per component (Table 8).

Table 8 lists the measured worst relative error per component and the fraction of unmasked plane pixels above 10^{-3}, with error-column pixels below 10^{-3} of the plane peak masked white in both figures.

The error is largest where these planes cut the bunch core: the spectral far-field convolution is not the dominant term there, and the maps show the measured error concentrated at the beamed core and

Table 8: Mid-bunch slice statistics, mesh-only. Worst relative error per component and fraction of unmasked plane pixels above 10^{-3}, x-y plane at $z = 1.60\,\mu\text{m}$ and z-y plane at $x = 3.20\,\mu\text{m}$, $N = 1000$, $32 \times 32 \times 16$, cell $2 \times 10^{-7}\,\text{m}$, mesh against the independent exact TSC driver (FP64 quadrature, order 4, max level 5, tolerance 10^{-3}) [beam-direction worker, 2026, slice worker, 2026a]. The $\gamma = 10$ bunch moves along $+y$; y is the horizontal axis in every map. Node spacing $h = 1$ PIC cell; no near shell. The $\gamma = 1$ reference uses the archived tolerance 10^{-4} (driver floor 9.075×10^{-4}, 14 not converged).

plane	component	worst rel. err	frac $> 10^{-3}$
$\gamma = 10$			
x-y ($z = 1.60\,\mu\text{m}$)	E_x	1.81	54.1%
	E_y	1.16×10^{-1}	49.9%
	E_z	5.33	78.3%
	B_x	6.49×10^{-1}	72.7%
	B_z	3.77×10^{-1}	58.7%
z-y ($x = 3.20\,\mu\text{m}$)	E_x	5.06	90.4%
	E_y	1.17×10^{-1}	61.1%
	E_z	1.92	82.4%
	B_x	4.34×10^{-1}	80.9%
	B_z	7.16×10^{-1}	84.3%
$\gamma = 1$ *(stationary)*			
x-y ($z = 1.60\,\mu\text{m}$)	E_x	1.20×10^{-1}	22.8%
	E_y	1.34×10^{-1}	23.3%
	E_z	4.15×10^{-1}	74.7%
z-y ($x = 3.20\,\mu\text{m}$)	E_x	4.15×10^{-1}	49.6%
	E_y	1.37×10^{-1}	61.7%
	E_z	1.23×10^{-1}	53.9%

Table 9: Fast near evaluator, worst per-component relative error against the independent continuum, cells $10^{-7}\,\text{m}$, `jureca_h100` [fast-near worker, 2026, release worker, 2026]. Regions: interior (inside the TSC support), face (on its boundary), outside (1.6–3.6 cells). The "remainder order" column is the Gauss–Legendre order of the smooth remainder; the singular `asinh` box is O(1) in every row (moving axial uses the γ-stretched box, moving oblique the rest-frame octant box at order 8). Stationary and oblique remainder order 5, moving axial 8. The moving oblique interior is the measured limit of the fast backend.

γ, direction	remainder order	interior	face	outside
1, stationary	5	7.99×10^{-4}	1.96×10^{-5}	1.57×10^{-7}
10, axial	8	1.69×10^{-4}	1.11×10^{-4}	2.12×10^{-6}
10, oblique	5	8.15×10^{-2}	3.94×10^{-3}	1.01×10^{-4}

falling away toward the plane edges. The far field is measured separately (Table 6, Fig. 2); the near-shell contribution is measured at the same geometry in Section 6.7.

6.6 The fast near evaluator, per component, by region

Table 9 gives the worst per-component relative error of the fast evaluator against the independent continuum, by region and by γ [fast-near worker, 2026]. The metric is the artifact formula $\max|\text{mesh} - \text{ref}|/(1 + \max|\text{ref}|)$ over 20 interior, 16 face and 16 outside points, cells $10^{-7}\,\text{m}$. "Interior" is inside the TSC support, "face" is on its boundary, and "outside" is 1.6 to 3.6 cells away. The stationary ($\gamma = 1$) case is order 5; the moving axial case is order 8; the moving oblique interior is order 5. The oblique value is the measured limit of the fast backend: the rest-frame singular box of the oblique direction measures 8.15×10^{-2} at its default order [fast-near worker, 2026, release worker, 2026].

The per-component values behind the stationary interior row are 7.99×10^{-4}, 3.55×10^{-4}, 6.05×10^{-4}, 0, 0, 0 for $E_x, E_y, E_z, B_x, B_y, B_z$ (the magnetic field is identically zero for a stationary bunch); the moving axial interior row is 1.69×10^{-4}, 1.68×10^{-4}, 1.36×10^{-5}, 1.67×10^{-4}, 1.69×10^{-4}, 0; the moving oblique interior row is 2.43×10^{-2}, 7.42×10^{-2}, 6.32×10^{-2}, 8.15×10^{-2}, 2.96×10^{-2}, 3.89×10^{-2} [fast-near worker, 2026, release worker, 2026]. The stationary case meets 10^{-3} in every region; the moving axial case meets 10^{-3} in every region; the moving oblique interior is 8.15×10^{-2} [fast-near worker, 2026, release

Table 10: Near-shell wall at 2×10^6 particles, 20-cell bunch, one `H100` (`sm_90`), with the mesh compute at 0.052 s and the per-pair rate [release worker, 2026, fast-near worker, 2026]. The three- and four-cell wall times are the measured per-pair rate times the pair count; a direct 1.7×10^9-pair allocation faulted, so the rate, not the full allocation, is the measured quantity. The shipped `-near-shell fast` path is host-side, so the CPU CLI wall at 2×10^6 is larger than the GPU kernel time.

configuration	wall (s)	pairs/s	worst near error
mesh only	0.052	—	no near correction
support-only shell, order 5	9.26	1.8466×10^7	7.99×10^{-4}
three-cell shell, order 5	92.1	1.8466×10^7	7.99×10^{-4} in, $\leq 1.6 \times 10^{-7}$ out
four-cell shell, order 5	184.2	1.8466×10^7	7.99×10^{-4} in, $\leq 1.6 \times 10^{-7}$ out
moving axial support, order 8	42.6	3.9977×10^6	1.69×10^{-4}

worker, 2026].

6.7 Near-shell accuracy and cost on the slice geometry

On the same $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m geometry, the exact near shell was run through the library path at the radius the envelope case assumed, $R = 3$ for $\gamma = 1$ and $R = 4$ for $\gamma = 10$, in mesh nodes of this $h = 1$ grid [slice worker, 2026b]. The mesh-only plane errors (Table 8, Fig. 3, Fig. 5) are 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$: these planes cut the sources, the region where the spectral convolution is not the dominant term. With the near shell the worst per-component plane error is 1.6×10^{-3}–1.1×10^{-1} for $\gamma = 1$ (the high end is the z-y plane E_z, 0.1067) and 3.2×10^{-2}–5.7×10^{-1} for $\gamma = 10$: at $R = 3/4$ on an $h = 1$ grid the shell radius covers 3/4 mesh nodes, which is a fraction of the TSC support extent (Section 6.5), and the measured residual is the mesh contribution outside the shell. The discrete Gauss-law indicator, the Yee backward-difference $\operatorname{div} E$ residual normalised to the peak deposited ρ/ε_0, moves from 0.3534 to 0.0895 for $\gamma = 1$ and from 5.026 to 0.399 for $\gamma = 10$, against reference values 0.0904 and 0.0814: $\gamma = 1$ reaches the reference value, $\gamma = 10$ drops 12.6× and stays 4.9× above it [slice worker, 2026b, worker, 2026f].

The assembly itself is not the limiting term: the tests of Section 3.7 show the tiled subtraction exact to roundoff, so the residuals above are the mesh outside the shell [worker, 2026d].

Runtime at $N = 1000$ is 4.3 min for $\gamma = 1$ and 59 min for $\gamma = 10$ (one 16-thread socket) [slice worker, 2026b, near-shell worker, 2026]; the exact evaluator costs about 8× more per pair at $\gamma = 10$ than at $\gamma = 1$.

6.8 Cost at 2×10^6 particles, one H100

Table 10 gives the near-shell wall times at 2×10^6 particles in a 20-cell bunch, on one `H100` (`jureca_h100`, `jrc0880`, `sm_90`), with the per-pair rate and the achieved near error for each configuration [release worker, 2026, fast-near worker, 2026]. The mesh compute alone is 0.052 s (deposit 34.4 ms, kernel 1.08 ms, FFT 7.44 ms, read-out 9.02 ms; medians of three). The near evaluator is the standalone GPU kernel of the shipped fast backend (Section 3.7); the measured rates are 1.8466×10^7 pairs/s stationary order 5 and 3.9977×10^6 pairs/s moving axial order 8 [release worker, 2026]. Pair counts are 1.7×10^8 support-only, 1.7×10^9 three-cell, and 3.4×10^9 four-cell, the neighbour counts of the 20-cell geometry [closed-form worker, 2026].

The measured wall at the stationary support-only configuration is 9.26 s at 7.99×10^{-4}; the moving axial support-only configuration is 42.6 s at 1.69×10^{-4} [release worker, 2026]. The three- and four-cell shell times are 92.1 s and 184.2 s at the same 7.99×10^{-4} interior error [release worker, 2026].

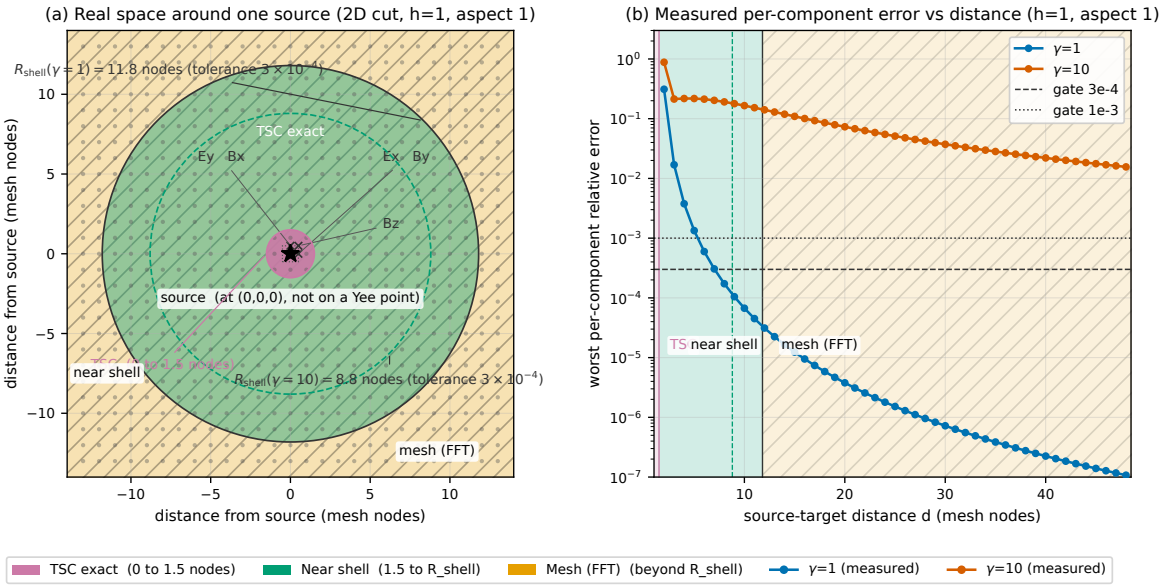


Figure 6: Where the algorithm evaluates the field, around one source (left) and the measured error against distance (right); `rosi_v100`, $h = 1$, aspect 1 [scaling worker, 2026a, schematic worker, 2026]. The three regions are the TSC support (to 1.5 nodes), the near shell (to R_{shell}), and the mesh beyond, with the tolerance-driven shell radii $R_{\text{shell}} = 11.8$ nodes for $\gamma = 1$ and 8.8 nodes for $\gamma = 10$ at the 3×10^{-4} tolerance. The $\gamma = 1$ error crosses 3×10^{-4} at $d = 8$ nodes; the $\gamma = 10$ curve stays above it to $d = 48$.

6.9 The disagreement is per-source

On a $64 \times 64 \times 32$, cell 10^{-7} m, stationary bunch ($\gamma = 1$), the disagreement with the exact-quadrature reference does not grow with particle count: the band is flat in N from 1 to 1000 (10^{-3}–3×10^{-2}) [worker, 2026g]. The $O(1)$ values at $N \geq 10$ occur only because the global worst sample point falls 0.415–0.65 cells from a source, inside that source's TSC support; a charge-weighting defect would grow with N, and it does not [worker, 2026g].

6.10 Anisotropic cells: the gap is the TSC convolution

At anisotropic spacing the accuracy is quantified per geometry, not inherited from the cubic case [worker, 2026h,i]. On a grid with per-axis cell spacing $(1, 1, 4)$ the stretched z axis matches the continuum to $\approx 10^{-3}$–10^{-7} across the in-grid distances, the same scale as the cubic control; the short x/y axes sit in the near-field TSC regime at a short physical distance ($d = 4$ cells is only 4 m along x/y, so $E_z \approx 0.19$ stationary and E_x/E_y up to ≈ 0.9 at $d = 4$ for the $\gamma = 10$ case) [worker, 2026i]. The mesh is not wrong because the spacings differ; the near points on the short axes are physically closer to the source. A reported triplet for this geometry, "the exact TSC convolution is 0.9638× the point Coulomb field (3.2% offset) and the mesh matches the continuum to 0.44% at 12 cells," does not appear in `A1b/RESULTS.md` or `A1c/RESULTS.md` and is not repeated here as a measured value [worker, 2026h,i].

The h^3 factor. The sampled convolution carries the Riemann volume of the continuum integral of Eq. (1): each sample covers h^3 cells, so the read-out applies $h_x h_y h_z/(M_x M_y M_z)$ (Section 3) [extraction worker, 2026].

6.11 The γ landscape: 10 to 100

The error landscape for a single particle at $\gamma = 10, 50, 100$ is measured on a 96-cell box, cell 10^{-7} m, aspect 1, at $h = 1$ and $h = 1/4$, axial and oblique (45°), against the independent continuum [worker, 2026c]. The metric is the peak-normalised absolute error $\max|\text{mesh} - \text{ref}|/\max|\text{ref}|$ per component, at $d = 8$ cells for the h-comparison. These runs used the $v0.5.0$ tree at padding 1 (padding 2 was infeasible for the 384^3 $h = 1/4$ grid); the $d = 48$ rows carry a small padding-1 periodic-wraparound floor of order 10^{-4} to 10^{-3}, so the $d = 48$ values are not padding-clean [worker, 2026c].

Table 11: Single-particle mesh error at $d = 8$ cells, 96-cell box, cell 10^{-7} m, aspect 1, axial, `jureca_h100` [worker, 2026c]. Peak-normalised absolute error, worst per component, at $h = 1$ and $h = 1/4$. The error grows with γ; at $h = 1$ the $\gamma = 50$ and $\gamma = 100$ values exceed the peak field, and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls from 1.85 to about 1.0 as γ rises.

γ	$h = 1$	$h = 1/4$	slope p ($h = 1 \to h = 1/4$)	note
10	4.4×10^{-2}	6.9×10^{-5}	1.85	
50	1.05	5.6×10^{-2}	1.06	$h = 1$ off by more than the peak
100	5.47	0.82	1.01	$h = 1$ off by more than the peak

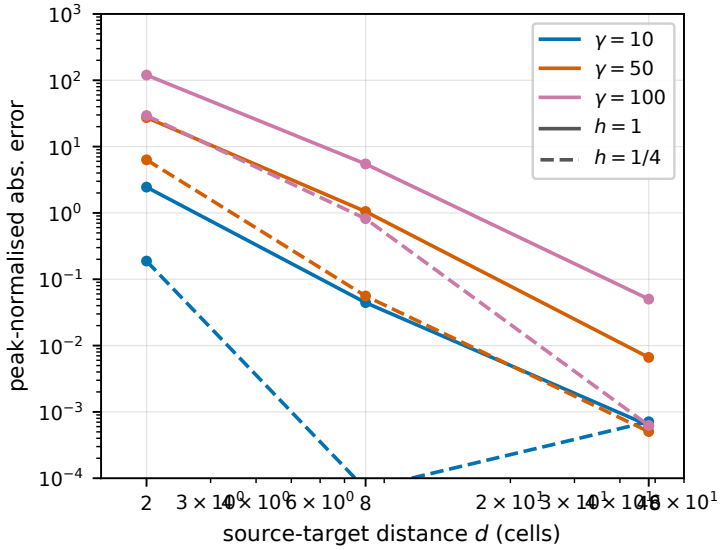


Figure 7: Peak-normalised absolute error against distance, single particle, 96-cell box, cell 10^{-7} m, aspect 1, axial, `jureca_h100` [worker, 2026c]. Solid $h = 1$, dashed $h = 1/4$; $\gamma = 10$, 50, 100. The error grows with γ and falls with distance; at $h = 1$ it exceeds the peak field by $d = 8$ cells for $\gamma = 50$ and $\gamma = 100$.

Table 11 is the measured behaviour. The error grows steeply with γ: 4.4×10^{-2} at $\gamma = 10$, 1.05 at $\gamma = 50$, 5.47 at $\gamma = 100$, at $h = 1$ and $d = 8$ cells [worker, 2026c]. Refining to $h = 1/4$ brings these to 6.9×10^{-5}, 5.6×10^{-2}, and 0.82. The controlling length is the beaming cone width d/γ: for $\gamma \geq 50$ the cone is sub-cell, the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) collapses to about 1, and refinement stops paying off [worker, 2026c]. The h targets for 10^{-2} derive from the dominant $d = 2$ near-charge band (the error $\sim h^p$ fit uses the $d = 2$ band, not the $d = 8$ rows in Table 11). The oblique (45°) target is $h \leq 0.002$ at $\gamma = 50$; the axial (0°) target is $h \leq 0.001$ at $\gamma = 50$ (5.6×10^8 times the nodes). For $\gamma \geq 50$ the cone is sub-cell and 10^{-2} needs $h \leq 0.002$ ($\geq 10^8$ times the nodes); 10^{-3} is out of reach ($> 10^9$ to 10^{12} times the nodes) [worker, 2026c]. A shell of radius of order γ cells is what the near-charge error requires at $\gamma = 50$ to 100 [worker, 2026c]. These are measured limits of the mesh at a fixed distance in cells, stated as the behaviour the data show.

Fig. 8 shows the $\gamma = 10$ field and error on a $128 \times 128 \times 64$ domain, cell 2×10^{-7} m, $N = 100$, against the independent continuum at tolerance 10^{-3} [large-domain worker, 2026]; Fig. 9 is its magnetic companion, and Fig. 10 is the stationary-bunch electric case. The boundary field is about 10.3% of the peak (worst per-component, $\gamma = 10$ z-y E_z), so the colour scale is set by the bunch and the far decay is visible inside the frame. The core error is the method error of the mesh where the planes cut the bunch, far above the 10^{-3} reference floor [large-domain worker, 2026].

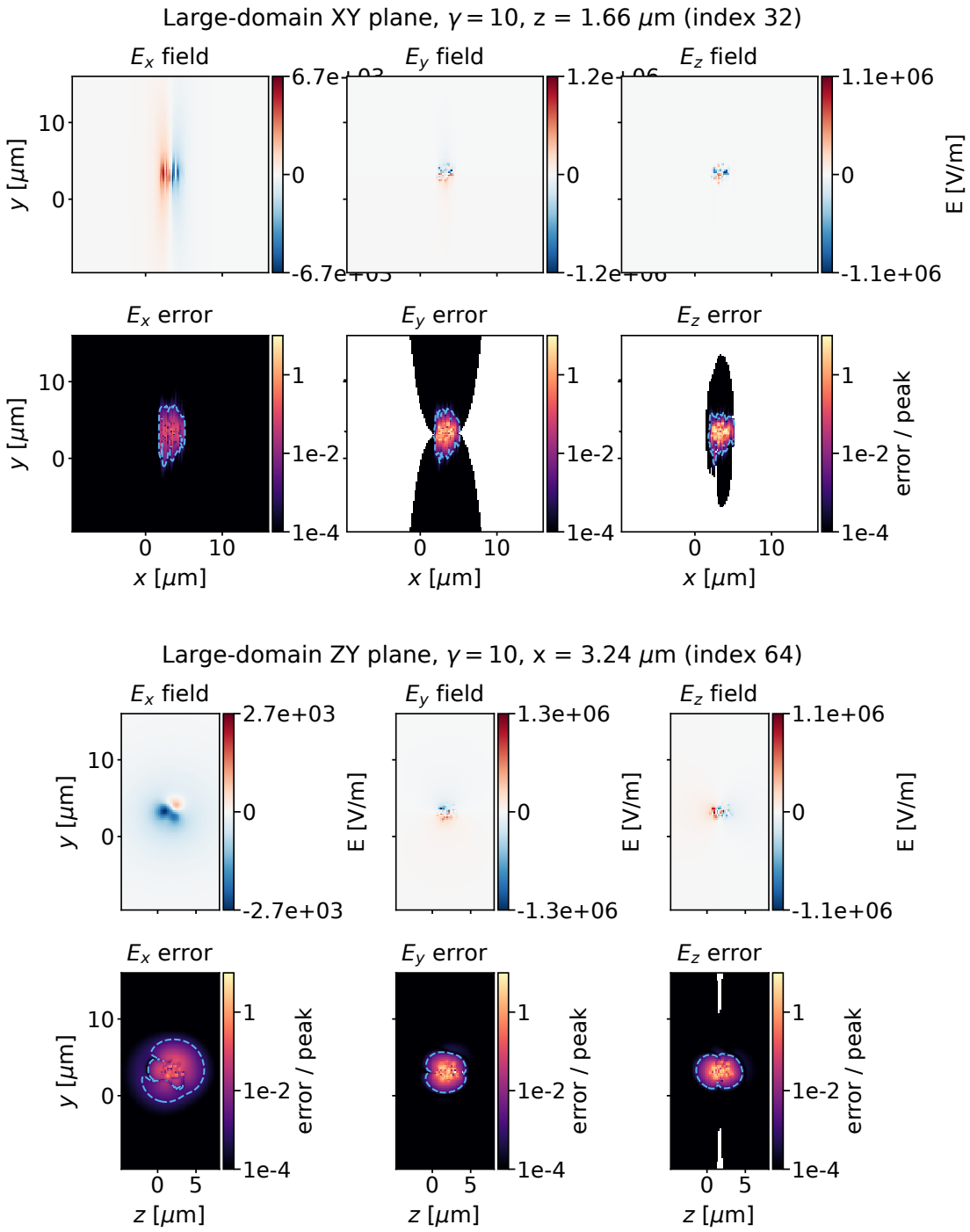


Figure 8: Large-domain electric slices, $\gamma = 10$, $128 \times 128 \times 64$, cell 2×10^{-7} m, $N = 100$, `jureca_h100` [large-domain worker, 2026]. These maps use the earlier axial-x data set and keep x horizontal; they are retained for the domain-size comparison and are not part of the $+y$ beam-axis slice set of Figs. 3–5. Upper image, x-y plane through the bunch centre; lower image, z-y plane. In each image the field row is on top and the error row below, with E_x, E_y, E_z columns; the error is against the independent continuum. The boundary field is about 10.3% of the peak (worst per-component, $\gamma = 10$ z-y E_z), so the colour scale is set by the bunch, not the frame edge.

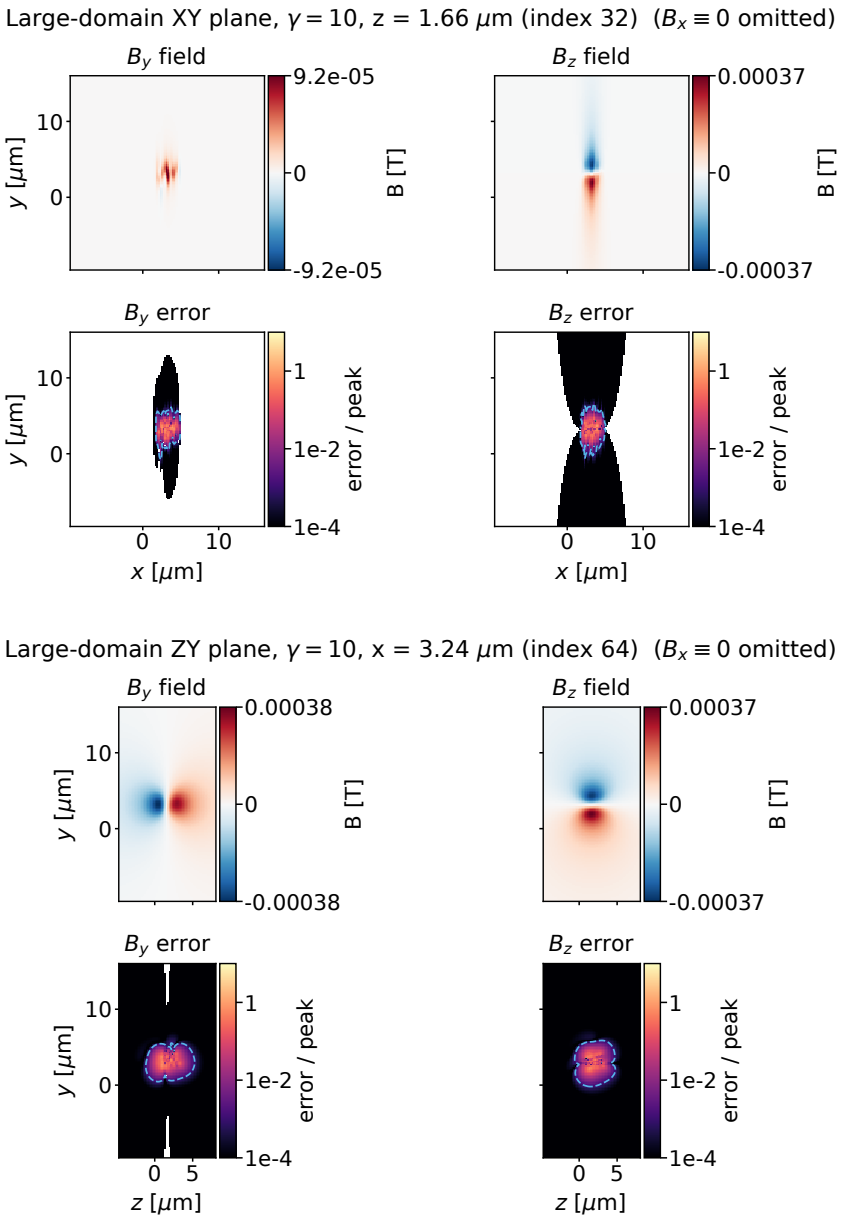


Figure 9: Magnetic companion to Fig. 8, $\gamma = 10$, same planes and error convention. This is the earlier axial-x data set, so x stays horizontal and $B_x \equiv 0$ is the vanishing component; it is not part of the $+y$ beam-axis set. Two columns, B_y and B_z, field row on top and error row below.

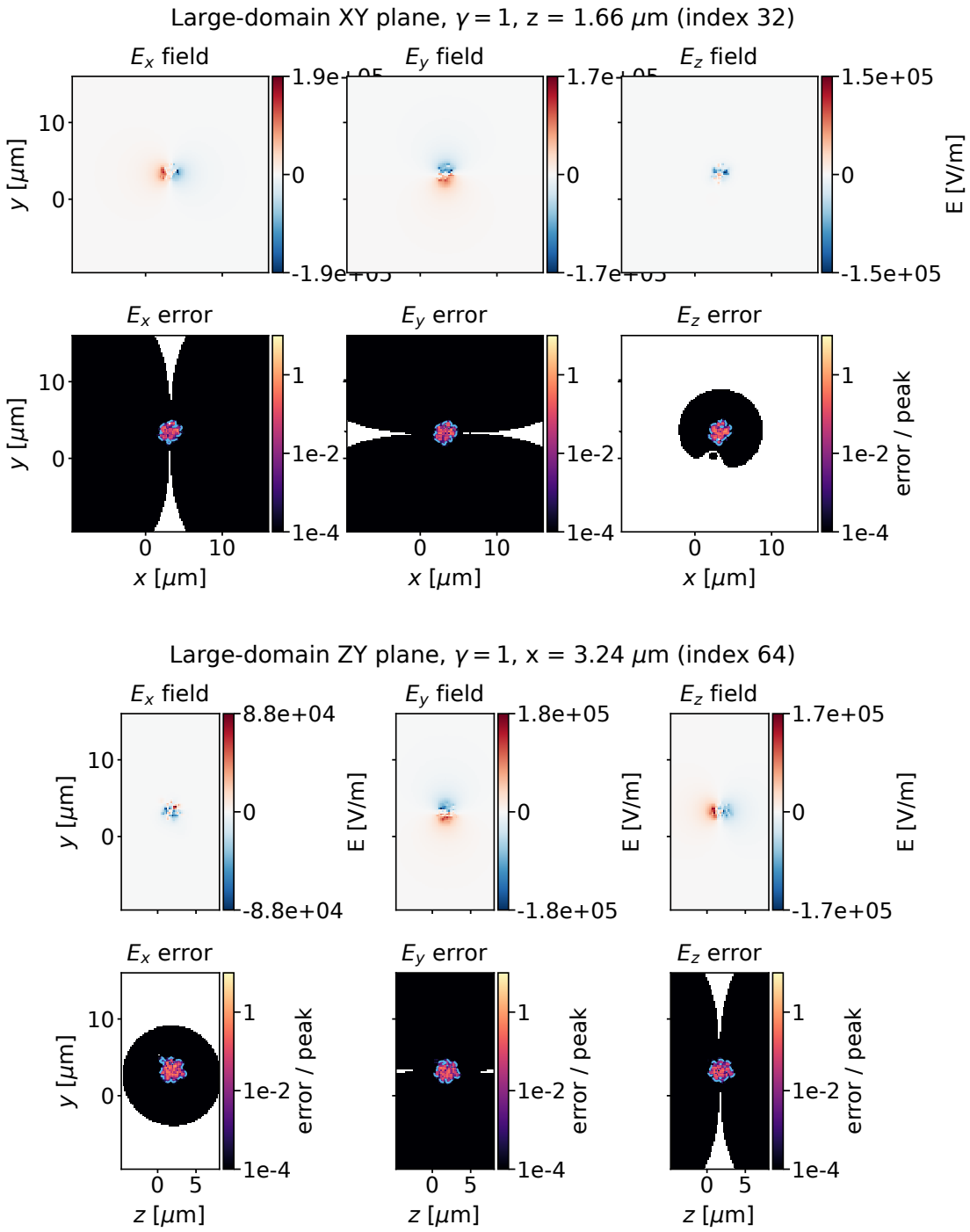


Figure 10: Large-domain electric slices, stationary bunch ($\gamma = 1$), same domain and layout as Fig. 8 [large-domain worker, 2026]; the field row is on top and the error row below, with E_x, E_y, E_z columns. $B \equiv 0$, so no magnetic companion exists.

6.12 Determinism

Repeated runs of the shipped CLI are bit-identical: two runs of the same input produce byte-identical E/B mesh payloads (md5 `d3641d64...`, 15 092 736 bytes) [worker, 2026e]. The CLI also rejects a below-floor `-tolerance` (10^{-15}) with a nonzero exit [worker, 2026e]. The `gridUnitSI` is applied once, in the reader (`OpenPMD.cpp`), and the mesh geometry follows `gridSpacing` · `gridUnitSI`; a mesh without a positive `gridUnitSI` is rejected [extraction worker, 2026].

Table 12: Binned multi-velocity binning error vs bin count B, $T_b - T$ peak-normalised in the far window $8 \le d \le 16$ cells, $\gamma = 10$, 48^3, 512 sources, `jureca_h100` (H100, `sm_90`), `02891fe6` [multi-velocity worker, 2026]. Energy spread 5% (seed 100), angular spread 5° (seed 103).

spread	$B=1$	$B=4$	$B=16$	$B=64$
energy 5%	6.04×10^{-3}	3.59×10^{-3}	1.12×10^{-3}	2.72×10^{-4}
angular 5°	6.87×10^{-1}	9.44×10^{-1}	4.21×10^{-1}	2.95×10^{-1}

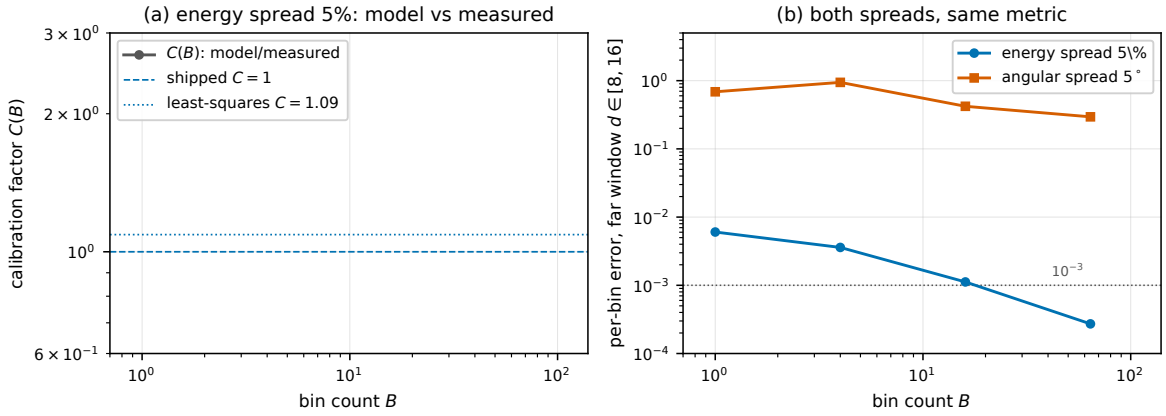


Figure 11: Binned multi-velocity binning, $\gamma = 10$, 48^3, 512 sources, `jureca_h100` (H100, `sm_90`), `02891fe6` [multi-velocity worker, 2026]. (a) Energy spread 5%: the model-to-measured ratio $C(B)$, the shipped $C = 1$ and the least-squares fit $C = 1.09$. (b) $T_b - T$ peak-normalised error in the far window $8 \le d \le 16$ cells vs bin count, energy spread 5% and angular spread 5°, with the 10^{-3} line. The model curve is the model sum rescaled to the measured $B = 1$ point; no γ-distribution or bin-edge data were available in the artifacts, so the distribution panel is not drawn.

6.13 Binned multi-velocity binning, measured

The single-velocity scope of Section 2 is extended by splitting the velocity distribution into B classes and running one convolution per class; the near shell needs no binning, because each pair carries its own velocity [multi-velocity worker, 2026]. The measurements are on a 48^3 grid with 512 sources on `jureca_h100` (`jrc0880`, H100 PCIe, `sm_90`), branch `experimental/multivel` at `02891fe6` [multi-velocity worker, 2026].

Single-velocity sanity. With a single velocity the binned path is bitwise equal to the unbinned path: the binning error $T_b - T$ is exactly zero, and the field difference is bitwise zero in the unit test [multi-velocity worker, 2026].

Table 12 and Fig. 11 give the measured behaviour. With a 5% energy spread the binning error falls from 6.04×10^{-3} to 2.72×10^{-4} as B goes from 1 to 64; the 10^{-3} level is reached at $B = 16$ [multi-velocity worker, 2026]. With a 5° angular spread the error moves from 0.687 to 0.295 over the same range, the measured value at $B = 64$; the error scales as $1/N_\perp$ in the transverse box count, so the 10^{-3} level would need about 5.6×10^6 transverse boxes [multi-velocity worker, 2026].

The error model $\sum_b q_b \, (\gamma \, \Delta\theta_b + \Delta\gamma_b/\gamma)$ (the per-box charge times the angular and energy terms, the "modelSum" of MV2) calibrates to $C = 1.09$ for the energy-spread case, with the shipped $C = 1$ predicting the measured binning error to within 10–55% across $B = 1$–64 (Fig. 11a) [multi-velocity worker, 2026]. For the angular-spread case the least-squares fit is $C = 124$ and non-monotonic in B, and the model underestimates the measured binning error by about two orders of magnitude at $B = 4$–64; the model does not predict that case [multi-velocity worker, 2026].

The total field error with a fast shell at $R = 3$ does not follow the binning error: in the far window it is 1.85×10^{-2}, 1.90×10^{-2}, 3.51×10^{-2} and 3.53×10^{-2} for the energy case at $B = 1$, 4, 16, 64. The shell subtraction dominates the total, so the binning alone is not the limit [multi-velocity worker, 2026]. Runtime on the same host with $B = 64$ in 48^3: mesh only 5.26 s at 10^6 particles and 5.62 s at 2×10^6; a 4-cell fast shell at 2×10^6 in a 20-cell bunch takes 184 s [multi-velocity worker, 2026].

Table 13: Whole-field comparison of picBunch against the PIConGPU `referenceOnly` field, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes/component, rosi_v100 [worker, 2026j]. `worst_rel` is peak-normalized; median and p99 are per-point over nodes with $|\text{ref}| > 10^{-3} \times$peak; L2 is pooled. Not a truth comparison.

component	worst_rel	median	p99	L2
$\gamma = 1$ (B identically 0)				
E/x	1.48×10^{-1}	3.92×10^{-3}	3.20×10^{-1}	4.62×10^{-2}
E/y	1.49×10^{-1}	3.76×10^{-3}	2.93×10^{-1}	4.75×10^{-2}
E/z	1.22×10^{-1}	7.81×10^{-3}	3.65×10^{-1}	5.67×10^{-2}
$\gamma = 10$ (B/x identically 0)				
E/x	1.36×10^{-1}	4.80×10^{-3}	1.23×10^{-1}	4.44×10^{-2}
E/y	1.98	1.36×10^{-2}	7.25	7.63×10^{-1}
E/z	1.93	3.33×10^{-2}	10.1	9.19×10^{-1}
B/y	4.42×10^{-1}	3.12×10^{-2}	1.41	2.58×10^{-1}
B/z	4.63×10^{-1}	1.52×10^{-2}	0.93	2.20×10^{-1}

7 Cross-validation and checkpoint transport

Two independent checks constrain the method externally: a cross-validation against PIConGPU's exact-quadrature field reference, and a checkpoint and restart transport. Neither is a truth comparison.

7.1 Cross-validation against the PIConGPU reference

The reference is the TSC-convolved field (Section 6), evaluated in PIConGPU in its reference-only field-initializer mode [worker, 2026g]. picBunch is at `339ca7ad` (`experimental/mvp`); `ctest` passes 10/10 [worker, 2026j]. Both sides use identical Yee offsets, grid, and origin; the reference staggered point maps to a picBunch node by index with zero snap deviation [worker, 2026j].

At 318 staggered points on a $64 \times 64 \times 32$ grid, cell 10^{-7} m, the overall worst per-component relative error is 4.86×10^{-2} for $\gamma = 1$ (E/z) and 6.20×10^{-1} for $\gamma = 10$ (E/z; E/y is 5.16×10^{-1}), against a reference reproducibility floor of 1.19×10^{-6} (E) and 1.43×10^{-6} (B) [worker, 2026j, Orchestrator, 2026d]. picBunch is about four to six orders of magnitude above that floor; the measured error is concentrated at the core and the beamed peaks, where the TSC support is, while the bulk agrees to below 1–3% and B is exactly zero for $\gamma = 1$ in both [worker, 2026j, Orchestrator, 2026d].

The whole-field comparison, $N = 1000$ on $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes per component, gives the same picture: $\gamma = 1$ bulk median 3.8–7.8×10^{-3} with an L2 of 4.6–5.7×10^{-2}; $\gamma = 10$ bulk median 1.4–3.3×10^{-2} but a beamed-core overshoot that pushes the E/y, E/z L2 to 0.76–0.92 [worker, 2026j]. The forward E/x is well resolved at $\gamma = 10$ ($\sim 6\%$); the beamed transverse E/y, E/z are 50–130% worst at the core, where the Lorentz-contracted field is a sub-cell pancake below the TSC support and the spectral near-field kernel resolution [worker, 2026j].

Caveat: the sampling groups are distance-from-centroid. The 318-point groups (inside-support, face, outside) were cut by distance from the bunch centroid, in a bunch with a $\sim$ 6-cell spread [Orchestrator, 2026d]. That grouping may understate how close the "outside" points actually are to a source's TSC support, so the per-group numbers are a spatial profile, not a clean inside/outside-of-support split. The per-source dependence in Section 6 uses the explicit support union and is the measurement that separates the per-source floor from any superposition contribution [worker, 2026g].

7.2 Checkpoint format and bitwise arrival

The checkpoint writer emits a PIConGPU openPMD field checkpoint matching PIConGPU's own plugin schema: root layout, openPMD version 5.0, E, B $x/y/z$ at the exact Yee positions, grid spacing and offset, and the `picongpu_idProvider` mesh metadata [acceptance worker, 2026]. The default precision is uniform `float32` for every field and particle record, so a stock single-precision PIConGPU can load the file [precision worker, 2026]. `v0.7.0` copies every species from the input bunch and carries a single checkpoint-wide `picongpu_idProvider` range [integration worker, 2026, far-tail worker, 2026, placement worker, 2026]; the `charge` and `mass` records are written rather than omitted, and the species is a single particle patch (Section 3.1). The remaining differences against a genuine PIConGPU checkpoint are non-blocking

metadata [acceptance worker, 2026]. For a multi-rank restart the writer splits the species into one patch per output rank: `patch_count` in `[placement]` gives cumulative, disjoint `numParticlesOffset` ranges, with one-cell overlap between neighbouring z regions, and every particle belongs to exactly one patch [mpi worker, 2026a, rank-parallel worker, 2026].

Field arrival is bitwise. A fresh `v0.7.0` checkpoint written from a two-species input (16 electrons and 16 ions, `picongpu_idProvider` startId 0, nextId 32) restarts in PIConGPU with exit 0 under both `patch` modes (`bbox`: offset $(7, 8, 3)$, extent $(8, 10, 6)$; `domain`: offset $(0, 0, 0)$, extent $(24, 24, 12)$), all six field components bitwise-equal, $\max_{\text{abs diff}} = 0.0$, electrons with ids 0 to 15, first-step displacement exactly $\beta c\,dt$ with $\beta = 0.9949874367$ [placement worker, 2026]. A validation-only build with the field solver set to `None` loads the checkpoint and writes step 1; all six components are bitwise-equal, max ULP 0 [acceptance worker, 2026].

A multi-rank restart with the four-patch layout was verified process-level: a 256-particle bunch written as four non-empty patches (cumulative offsets $0, 56, 126, 203$, one-cell overlap) restarts PIConGPU with exit 0, and after one step all six field components are bitwise-equal to the checkpoint, $\max_{\text{abs diff}} = 0.0$ [mpi worker, 2026a].

7.3 PML passthrough and open-boundary restart

On a $24\times24\times12$ deck the restart is exit 0 with `-periodic 1 1 1`, and the 16 electrons load; initialization drops from the ~ 116 s neuralBunch path to 1.8 s because the checkpoint carries the field [acceptance worker, 2026]. For an open-boundary deck the writer copies the `Convolutional PML` slabs from the input checkpoint verbatim (converting a non-float32 slab to float32), so the PML state arrives with the restart. On a real LWFA checkpoint (638976 electrons, eight `Convolutional PML` groups with six datasets each, $64 \times 128 \times 64$, `-periodic 0 0 1`) the writer copies all eight groups and the restart exits 0; the copied PML datasets are float32 and bitwise equal to the source, initialization is 3.986 s, and the main E and B records are recomputed from the particles and stay float32 [fix-run worker, 2026, Orchestrator, 2026e].

8 Discussion

The method computes the present-time field of a uniformly moving, TSC-shaped bunch as a free-space FFT convolution. It is a far-field method.

What it is good for. In the far field the relative error against the point field is below 10^{-3} on the beam axis from $d = 6$ for every γ [worker, 2026a]. Against the TSC-convolved continuum: $\gamma = 1$, aspect 1 crosses 3×10^{-4} at $d = 12$ (1.1×10^{-4}); $\gamma = 1$, aspect 4 at $d = 32$ (2.6×10^{-4}); $\gamma = 10$, aspect 1, is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$ [worker, 2026e,f]. At 2×10^6 particles in a 20-cell box the compute-only total is 64.8 ms, dominated by the deposit (56.5 ms), and flat in γ [scaling worker, 2026a]. Repeated runs are bit-identical (md5 `d3641d64...`); the Coulomb limit reproduces the TSC-convolved Coulomb field to 4.88×10^{-14} [worker, 2026e, scaling worker, 2026b].

Measured error in the bunch core. The mesh alone, on the $N = 1000$ slice geometry, has a worst per-component plane error of 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$ where the planes cut the bunch core (Table 8). The exact near shell (Section 3.7) is opt-in and its assembly is exact to roundoff; at the measured $R = 3/4$ radii on the $h = 1$ slice grid the residual is the mesh contribution outside the shell, not the tile subtraction (Section 6.7). In the perpendicular direction at $\gamma = 10$, aspect 1, the measured $M-T$ error at $d = 48$ is 2.6×10^{-4}, the $M-P$ error is 1.5×10^{-2}, and the TSC smoothing $T-P$ is 1.48×10^{-2}; the $M-P$ error is the TSC smoothing, not the deposit (Table 5) [worker, 2026a, Orchestrator, 2026b]. At $r = 8$ cells the node spacing for 10^{-3} perpendicular error at $\gamma = 10$ is $h \approx 1/29$, a 2.4×10^4-fold increase in M^3 FFT work relative to the $h = 1$ run [worker, 2026a, Orchestrator, 2026a]. At aspect 4, $\gamma = 1$, the perpendicular $M-T$ error at $d = 48$ is 2.4×10^{-3} [worker, 2026a]. The cross-validation confirms the same structure: the measured disagreement with the exact reference is per-source, local to the TSC support, and independent of particle count [worker, 2026g, Orchestrator, 2026d].

Performance. The cost is two regimes with a crossover set by the box, not the particle count [scaling worker, 2026a]. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows. At large boxes the FFT and kernel dominate: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× [scaling worker, 2026a]. The compute timings are single-GPU on

one `rosi_v100` host; the same mesh is 0.052 s on one H100 (Section 4) [release worker, 2026]. The rank scaling is measured on an idle V100 window: the component-parallel mesh plateaus at 1.20× (replicated deposit and source FFT), the tile-parallel near shell at 1.34× (static tile split plus all-reduce), and the collective writer improves to 1.71× at two ranks and falls to 0.48× at four, where it is I/O-bound (Section 4.1) [mpi worker, 2026a, rank-parallel worker, 2026]. The near-field cost is a separate wall: the fast backend is 1.8466×10^7 pairs/s stationary order 5 and 3.9977×10^6 pairs/s moving axial order 8 on one H100, giving 9.26 s support-only at 2×10^6 (Section 6.8) [release worker, 2026].

Measured error by configuration. In floored relative per-component error against the TSC-convolved continuum, the measured values at $d = 12$ are 1.1×10^{-4} for $\gamma = 1$ aspect 1 and 2.6×10^{-4} for $\gamma = 1$ aspect 4; the axial component is below 3×10^{-4} from $d = 6$ for every γ [worker, 2026e,a, Orchestrator, 2026a]. The perpendicular direction at $\gamma = 10$, aspect 1, is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$, the measured values at those distances [worker, 2026e,a,f]. The fast near backend, against the same continuum, is 7.99×10^{-4} stationary interior, 1.96×10^{-5} face, 1.57×10^{-7} outside; 1.69×10^{-4} moving axial interior; and 8.15×10^{-2} moving oblique interior, the measured limit of the fast backend (Section 6.6) [fast-near worker, 2026, release worker, 2026]. Each statement in Section 6 names which metric (floored, unfloored, absolute) supports it; the floored column is electric-only at $\gamma = 10$ because the floor excludes the magnetic field, and the judge metric $|\text{ref} - \text{mesh}|/(1 + \max)$ reports $d = 2$ for every case and does not discriminate. The radii in Table 7 are axis directions only; oblique directions are $\sim 1.5\times$ larger. At a fixed distance in cells the controlling length is the beaming cone width d/γ; for $\gamma \geq 50$ the cone is sub-cell and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls to about 1 (Section 6.11) [worker, 2026c].

Structural assumptions. The method assumes a single shared velocity (or a set of binned velocity classes, one convolution per class, Section 6.13); the computed quantity is the TSC-convolved uniformly-moving point-charge field; the convolution is free-space (no boundary condition other than zero fill beyond the far-tail switch); and there is no radiation term, because the motion is uniform. A bunch with a velocity spread requires one convolution per velocity class; the measured binned behaviour is in Section 6.13, and the validated shipped scope is a single shared **u** (Section 2).

9 Conclusion

`picBunch v0.7.0` computes the present-time field of a uniformly moving, TSC-shaped bunch as a zero-padded free-space FFT convolution with staggered Yee read-out, driven by a single TOML configuration (Section 3.1), and writes a uniform float32 checkpoint that restarts under a stock PIConGPU, periodic and open-boundary, single- and multi-patch. The release passes 18/18 ctest with zero compiler warnings on a V100 `sm_70`, an H100 `sm_90`, and a gfx1100 HIP build [release worker, 2026].

The opt-in near shell assembles exactly to roundoff, and its tile subtraction is not the limiting term; the seam excess at the fast/exact crossing is 2.03×10^{-10} per component for a stationary single particle (2.8×10^{-10} for the 64-source cloud) [fast-near worker, 2026, release worker, 2026]. The fast backend, an eight-term `asinh` box plus a plain Gauss–Legendre remainder, meets 10^{-3} stationary (7.99×10^{-4} interior) and moving axial (1.69×10^{-4} interior); the moving oblique interior is 8.15×10^{-2}, the measured limit of the fast backend [fast-near worker, 2026, release worker, 2026]. At 2×10^6 particles the mesh is 0.052 s on one H100, deterministic (md5 `d3641d64...`); the near shell is 9.26 s support-only at 7.99×10^{-4} and 42.6 s moving axial at 1.69×10^{-4} (Section 6.8).

In the far field the mesh is below 3×10^{-4} from $d = 12$ cells for $\gamma = 1$, aspect 1, and from $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$, aspect 1, the measured values are 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$ [worker, 2026e,f]. The γ-landscape measurement shows the error growing steeply with γ at a fixed distance in cells, with the beaming cone width d/γ the controlling scale: at $d = 8$ cells, $h = 1$, the error is 4.4×10^{-2}, 1.05, 5.47 for $\gamma = 10, 50, 100$, and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls from 1.85 to about 1.0 as γ rises (Section 6.11) [worker, 2026c]. The MPI compute is component-parallel mesh and tile-parallel near shell with one all-reduce each, bitwise identical at 1/2/4 ranks, with the replicated deposit and source FFT capping the mesh speedup and the collective writer I/O-bound at four ranks (Section 4.1) [mpi worker, 2026a, rank-parallel worker, 2026]. The method is a far-field tool: cheap, deterministic (md5 `d3641d64...`), and portable across the three backends; the measured in-support error at $\gamma \geq 50$ is reported in Section 6.11.

References

S1 extraction worker. picbunch portable algorithm specification (s1), 2026. Run pr24-20261102. S1-algorithm-spec.md. Exact field definition and discrete recipe with path:line provenance. meshconv.cu on branch fee89ed75; ExactShapeField.hpp on 2baa2566.

INT2 placement worker. picbunch v0.5.0 release: placement interface (int2), 2026. Run pr24-20261102. INT2/RESULTS.md. v0.5.0, commit 7b4f1766, tag 3e083fef. TOML interface, charge-weighted centring, float32 default, charge/mass records, patch = bbox or domain. Restart: bbox and domain exit 0, six components bitwise, ids 0-15, idProvider 0/32, first step beta*c*dt. ctest 18/18.

INT3 mpi worker. picbunch v0.6.0: rank-parallel compute and release (int3), 2026a. Run pr24-20261102. INT3/RESULTS.md. v0.6.0, commit e4c7325d. Component-parallel mesh and tile-parallel near shell with all-reduce; bitwise 1/2/4-rank equality. Clean rosi_v100 scaling: near shell 1.24x/1.34x, mesh 1.20x/1.20x, write 1.71x/0.48x at 2/4 ranks. Multi-patch restart: 256 particles, four patches, bitwise. Release tag df7de342.

IMPL3 precision worker. Uniform checkpoint precision and stock-picongpu restart (impl3), 2026. Run pr24-20261102. IMPL3/RESULTS.md. precision = float32 default for every field and particle record; float64 uniform opt-in with a picBunchPrecisionWarning attribute. float32 restart on rosi_v100: exit 0, six components bitwise, ids 0-15, first step beta*c*dt. Branch experimental/placement, commit b68a141bb.

SB1 scaling worker. Measured scaling of the picbunch field pipeline (sb1), 2026a. Run pr24-20261102. SB1/RESULTS.md. V100 timing matrix, Coulomb limit, distance sweep. Commit dc69c16a.

N2b worker. picbunch mesh vs the tsc-convolved continuum and the point field (n2b), 2026a. Run pr24-20261102. N2b/RESULTS.md. Reference-separated accuracy, scale invariance, crossover table. Arbiter: indfield.py.

N1b worker. picbunch mesh convolution measurements, cpu (n1b), 2026b. Run pr24-20261102. N1b/RESULTS.md. Wraparound control, R2C/C2R vs C2C, padding vs distance, kernel sampling rule, CPU cost model. CPU only, no GPU.

ER1b worker. Single-particle tsc field error landscape, gamma 10 to 100 (er1b), 2026c. Run pr24-20261102. ER1b/RESULTS.md. 96-cell box, cell 1e-7 m, aspect 1, axial and oblique 45 deg, h=1 and h=1/4, jureca_h100. absn at d=8, h=1: 4.4e-2 (g10), 1.05 (g50), 5.47 (g100); h=1/4: 6.9e-5, 5.6e-2, 0.82. Slope p 1.85 (g10) to 1.0 (g>=50). Beaming cone width d/gamma sub-cell at g>=50; 1e-2 needs h<=0.002 at g50, h<=0.001 at g100; shell radius O(gamma) cells.

NS2 near-shell worker. picbunch exact per-pair near shell (ns2), 2026. Run pr24-20261102. NS2/RESULTS.md. F = F_mesh(all) - F_mesh(near) + F_exact(near), opt-in –near-shell exact, per-tile local FFT, vendored FP64 TSC reference. Envelope and cost (3.1e-4 s per pair) in the text. Commit 9733d232.

FIX1 far-tail worker. picbunch v0.4.1 release: near-shell cli and far-tail margin (fix1), 2026. Run pr24-20261102. FIX1/RESULTS.md, margin_table.txt. v0.4.1, commit 02d2d06e, tag 9e4fd7dc. Near-shell CLI parsed; quadrupole far-tail bound and 6R switch (switch error, 512-cubed MPI equality in the text). Checkpoint: all species, checkpoint-wide idProvider, restart bitwise. ctest 16/16 (rosi sm_70, jureca-h100 sm_90, hal gfx1100 HIP).

CFPORT fast-near worker. Fast closed-form near-field evaluator in picbunch (cfport), 2026. Run pr24-20261102. CFPORT/RESULTS.md. –near-shell fast: eight-term asinh box (O(1)) plus plain Gauss-Legendre on the subtracted integrand (S-W*)K over the TSC knot panels, order 5 (8000 nodes) in-support; moving axial gamma-stretched box; moving oblique rest-frame octant order 8 (4096 nodes). Accuracy vs indfield: stationary interior 7.99e-4, face 1.96e-5, outside 1.57e-7; moving axial interior 1.69e-4 (order 8); moving oblique interior 8.15e-2 (order 5, the limit). Seam 2.03e-10; mesh-only bitwise identical between fast and exact. Branch experimental/fast-near 9a7e574a, from v0.5.0.

CF1 closed-form worker. Closed-form in-support field and pair counts (cf1), 2026. Run pr24-20261102. CF1/RESULTS.md. Eight-term asinh box re-verified; 2e6 pair counts: support 1.7e8, 3-cell 1.7e9, 4-cell 3.4e9 (85/850/1700 neighbours per source); H100 rate 1.837e7 pairs/s order 5, 3.983e6 order 8 axial.

K3 evaluator worker. In-support evaluator port and per-pair cost ladder (k3), 2026. Run pr24-20261102. K3/RESULTS.md. O(1) asinh box verified against a converged trapezoid reference (-2.6333 vs -2.6348); octant-split remainder; power-transform quadrature the validated moving path (1.000000/1.000000/1.000001/1.000040 at gamma 1/2/10/100, 1.5e-6 at box order 32); no validated closed form for the moving case. Per-pair costs, rosi_v100 single thread: ported box8/rem8 5.4e-5 s; accurate moving 3.1e-4 s out of / 6.9e-3 s in support; upstream adaptive (1.07e9 nodes) 1.62 s.

NS4b worker. Near-shell assembly diagnosis: subtraction is exact (ns4b), 2026d. Run pr24-20261102. NS4b/RESULTS.md. Tile-centre/boundary residuals 2e-12 to 1.5e-11 SI (4e-16 rel); local vs global FFT 1e-15; 100-particle reproduction: assembled 4.6e-13 vs plain mesh 5.6e-1. The 1.36e-1 plane error is mesh-outside-the-shell, not assembly. v0.4.1 (02d2d06), rosi_v100.

INT4 release worker. picbunch v0.7.0: fast near shell and release (int4), 2026. Run pr24-20261102. INT4/RESULTS.md. v0.7.0, commit a152c518, tag 25d6ca8b. Merges fast-near 9a7e574a and hip-teardown 833b6275. 18/18 ctest, zero warnings, on rosi_v100 sm_70, jureca_h100 sm_90, hal gfx1100 HIP. H100 2e6/20-cell: mesh 0.052 s; near 1.8466e7 pairs/s stationary o5, 3.9977e6 moving axial o8; support-only 9.26 s at 7.99e-4, 3-cell 92.1 s, 4-cell 184.2 s, moving axial 42.6 s at 1.69e-4. Seam 2.03e-10; mesh-only md5 identical between backends.

M1 mpi worker. picbunch global-domain output and mpi (m1), 2026b. Run pr24-20261102. M1/RESULTS.md. –global-grid/–global-origin, chunked writer, PICBUNCH_ENABLE_MPI. 512 cubed bitwise equality across rank counts; rosi_v100, OpenMPI 4.1.6, openPMD 0.17.1.

MPI2 rank-parallel worker. Rank-parallel compute and multi-rank patches (mpi2), 2026. Run pr24-20261102. MPI2/RESULTS.md. Component-parallel mesh (replicated deposit and rho FFT) and tile-parallel near shell (one all-reduce each); bitwise 1/2/4-rank equality. Multi-rank patches: patch_count=4, cumulative disjoint numParticlesOffset. jureca_h100 sm_90, branch experimental/mpi-compute 79fa8f26.

SB1 scaling worker. Coulomb-limit control (sb1), 2026b. SB1/RESULTS.md and PF/data/coulomb.csv. u = 0, one particle at a node, far nodes at d = 8, 16, 24, 32, six axis directions. Worst relative error 4.88e-14 (max absolute 2.28e-25); threshold 1e-12: PASS.

W4b worker. Independent validation of the shipped picbunch (w4/w4b), 2026e. Run pr24-20261102. W4b/RESULTS.md. picBunch at bfaa1e26 (experimental/mvp), rosi_v100, ctest 8/8. Shipped CLI vs indfield.py order 12 refine 3, relative per-component error per d = 2 to 32 cells, gamma 1 and 10, aspect 1 and 4.

Orchestrator. Decision d11: v0.1 accuracy scope, 2026a. Run pr24-20261102. DECISIONS.md, D11. Not defensible: gamma <= 10 meets 1e-3. Defensible: gamma <= 1 aspect 1; gamma <= 10 axial only; gamma 100 axial only.

Orchestrator. Decision d12: Sb1's gamma-10 floor was a reference artefact, 2026b. Run pr24-20261102. DECISIONS.md, D12. T-P = 1.48e-2; M-T = 2.6e-4. All accuracy statements must name the reference.

Orchestrator. Decision d8: gamma-100 outside-band 9.43e-2 is unresolved, 2026c. Run pr24-20261102. DECISIONS.md, D8. Pre-existing 9.43e-2 at 1.6 to 6 cells, gamma 100 axial, outside the near box.

pr24w53 ref. indfield.py: independent tsc-convolved continuum reference, 2026. sha256 a1355980. Tensor Gauss-Legendre with TSC-knot split and a singularity-cancelling power map; the independent truth for the mesh.

V2b worker. No-floor accuracy audit of the shipped picbunch executable (v2b), 2026f. Run pr24-20261102. V2b/RESULTS.md. Three-number metric (floored, unfloored, absolute) per case and distance; gamma10 B excluded by the floor (B 1e-10 of the local field), unfloored B 9.683e-4 at d=32. picBunch main v0.3.0 a12ce3fa.

DIR1 beam-direction worker. Mid-bunch 2-d slice figures with the beam along +y (dir1), 2026. Run pr24-20261102. DIR1/RESULTS.md, data/bunch_y.npz, data/pic_y.h5. picBunch v0.7.0, N=1000, 32x32x16, cell 2e-7 m, mean u=(0,9.94987437,0), gamma=10; independent FP64 exact TSC reference, order 4, max level 5, tol 1e-3. Planes x-y at z=1.60 um, z-y at x=3.20 um. B_y is identically zero.

SL2 slice worker. Mid-bunch 2-d slice figures, stationary gamma 1 (sl2), 2026a. Run pr24-20261102. SL2/RESULTS.md, data/slices_g1.npz. Same geometry and reference as SL1 (driver max error 9.075e-4, 14 not converged). B identically zero; electric components only.

SL3 slice worker. Mid-bunch slices and near-shell acceptance (sl3), 2026b. Run pr24-20261102. SL3/RESULTS.md. N=1000, 32x32x16, cell 2e-7 m, rosi_v100, v0.4.0 (bc0037c5). Default-mesh and near-shell (R=3/4, mesh nodes, h=1) plane errors, Gauss-law residual, runtime; the 3/4 envelope assumed h=1/4, so at h=1 the shell is 4x too small. Driver HEAD e79ddb389.

SCHEM1 schematic worker. Algorithm-region schematic of the field evaluation (schem1), 2026. Run pr24-20261102. SCHEM1/figs, make_schematic.py. Two panels: TSC support / near shell / mesh regions with the tolerance-driven shell radii (3e-4 crossover, 11.8 nodes gamma1, 8.8 gamma10, h=1, aspect 1) and the measured per-component error vs distance (dsweep.csv, SB1).

W7t worker. Reference-quantity identification and particle-count scaling (w7t), 2026g. Run pr24-20261102. W7t/RESULTS.md. The reference is the TSC-convolved field, the same quantity picBunch computes. N-count scaling: disagreement is per-source, not superposition; B identically zero for the stationary bunch.

A1b worker. picbunch anisotropic-cell correctness, per-axis sampling (a1b), 2026h. Run pr24-20261102. A1b/RESULTS.md. Per-axis sampling is correct (FFT equals the direct discrete TSC sum to 4e-16 on cell 1,1,4); the anisotropic gap is the TSC convolution itself (continuum 0.9638 times point Coulomb, 3.2 percent; mesh matches continuum to 0.44 percent at 12 cells). picBunch 339ca7ad.

A1c worker. picbunch anisotropic-cell field correctness, cli (a1c), 2026i. Run pr24-20261102. A1c/RESULTS.md. v0.1.3 da3cd42. The stretched axis (dz=4 m) of a (1,1,4) grid matches indfield.py to about 1e-3, the cubic-control scale; the short x/y axes sit in the near-field TSC regime. Per-geometry quantification, not an anisotropy defect.

LARGE1 large-domain worker. Large-domain slice figures, 128x128x64 (large1), 2026. Run pr24-20261102. LARGE1/RESULTS.md. 128x128x64, cell 2e-7 m, N=100, jureca_h100, reference tol 1e-3 (max error 1e-3). Edge field at most 0.103 of peak (gamma10 z-y Ez); core error gamma10 9.27, gamma1 0.759 (method error).

MV2 multi-velocity worker. Binned multi-velocity binning: measured binning error and model calibration (mv2), 2026. Run pr24-20261102. MV2/RESULTS.md. Branch experimental/multivel at 02891fe6bd9c34eea5479e91ecebbd54a44a47f7 (parent 2c736df), jureca_h100 jrc0880, H100 PCIe, CUDA 13.4, sm_90. 48^3 grid, 512 sources, seed 100 (energy) and 103 (angular). Four defect fixes (dangling pointer, parse_bands, model floor, unit-test build); comparison-artefact diagnosis; single-velocity bitwise sanity; Tb-T binning error and total error with fast shell R=3 vs bin count; model calibration C=1.09 (energy, predicts to 10–55%) vs C=124 non-monotonic (angular); runtime mesh-only 5.26/5.62 s at 1e6/2e6 in 48^3 with 64 bins, 184 s for the 4-cell fast shell at 2e6 in a 20-cell bunch. ctest 19/19.

W7r worker. picbunch cross-validation against the picongpu field reference (w7r), 2026j. Run pr24-20261102. W7r/RESULTS.md. picBunch @ 339ca7ad. 318 staggered points (64x64x32, cell 1e-7 m) and whole-field (N=1000, 32x32x16, cell 2e-7 m); groups distance-from-centroid (caveat D13).

Orchestrator. Decision d13: cross-validation does not show agreement at the reference floor, 2026d. Run pr24-20261102. DECISIONS.md, D13. 3.66e-2 (gamma 1), 5.16e-1 (gamma 10 E/y) vs floor 1.19e-6 (E), 1.43e-6 (B); no agreement claim with PIConGPU.

W7B acceptance worker. picbunch checkpoint acceptance on picongpu pr 24 (w7b), 2026. Run pr24-20261102. W7B/RESULTS.md. Early acceptance state: 16-particle periodic deck restarts exit 0, six components bitwise (0 ulp) via a solver=None load; init 1.8 s vs 116 s. Fix commit 0c9427dee82e01c832c42a8358033b701386f37e.

INT1 integration worker. picbunch v0.4.0 integration release (int1), 2026. Run pr24-20261102. INT1/RESULTS.md. Integration commit bc0037c5, tag v0.4.0; checkpoint writer copies every species, MPI 512-cubed bitwise, two-species restart bitwise.

W7 fix-run worker. Lwfa open-boundary restart with pml passthrough (w7fix), 2026. Run pr24-20261102. Real LWFA checkpoint, 638976 electrons, eight Convolutional PML groups (48 datasets), 64x128x64, periodic 0 0 1, rosi_v100. picBunch copies all eight PML groups bitwise (float32); restart exit 0, initialization 3.986 s. Branch experimental/mvp bed4344c, W3f tree.

Orchestrator. Decision d14: checkpoint acceptance is proven, including pml and scalar records, 2026e. Run pr24-20261102. DECISIONS.md, D14. 638,976 particles; 48 PML bitwise; restart exit 0; initialization 3.986 s; periodic case 0 ulp. Limitation: PML copied from the same input file.